

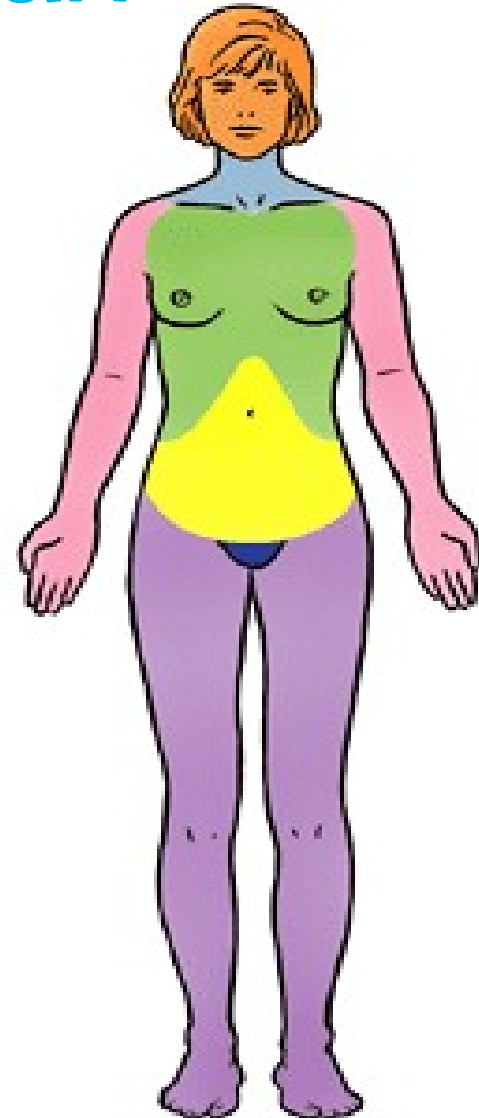
The Thorax

is the superior part of the trunk between the neck and abdomen.

Is divided into two parts for the descriptive purpose:

- The thoracic wall

- The thoracic cavity



Thorax

The thoracic wall consists of: skin, fascia, nerves, vessels, muscles, and thoracic skeleton.

The thoracic cavity- surrounded by the thoracic wall, contains the heart, lungs, thymus, distal part of the trachea, and most of the esophagus.

The Thorax

Functions of the thoracic wall include:

- protecting of thoracic & abdominal internal organs;
- Resisting negative internal pressures generated by elastic recoil of the lungs & inspiratory movements;
- providing attachment for and supporting the weight of the upper limbs.
- It provides attachment for muscles of upper limbs, neck, abdomen & back, & muscles of respiration.

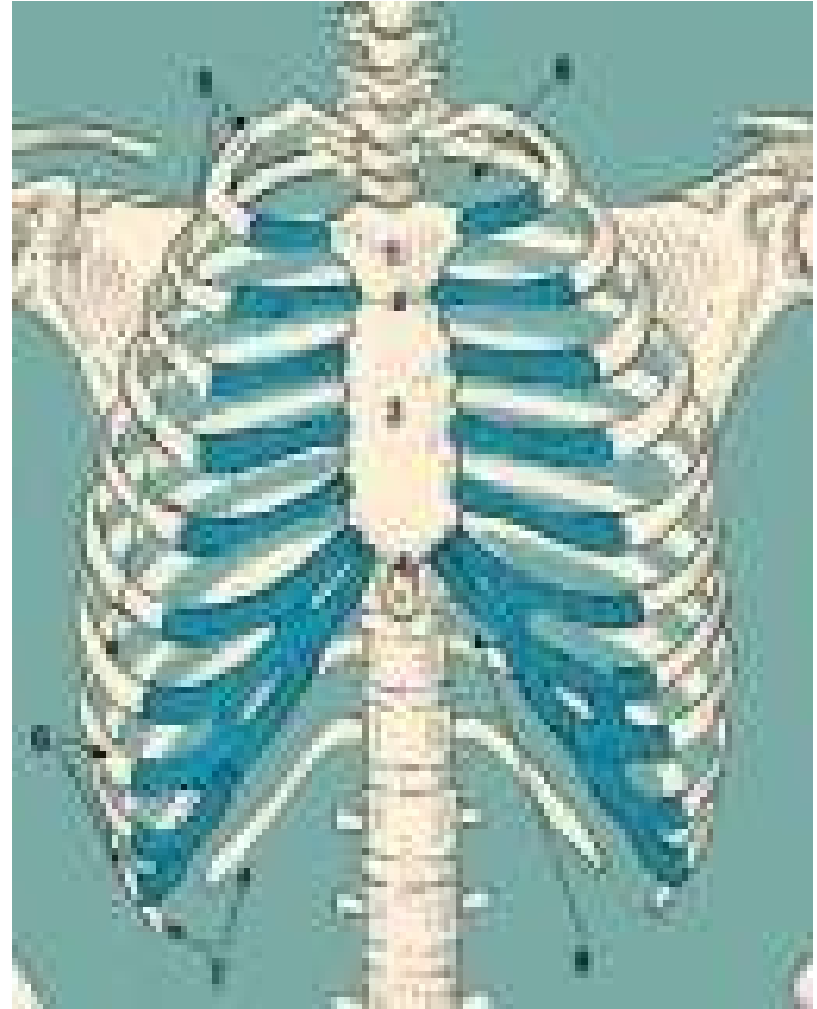
The Thorax: bones

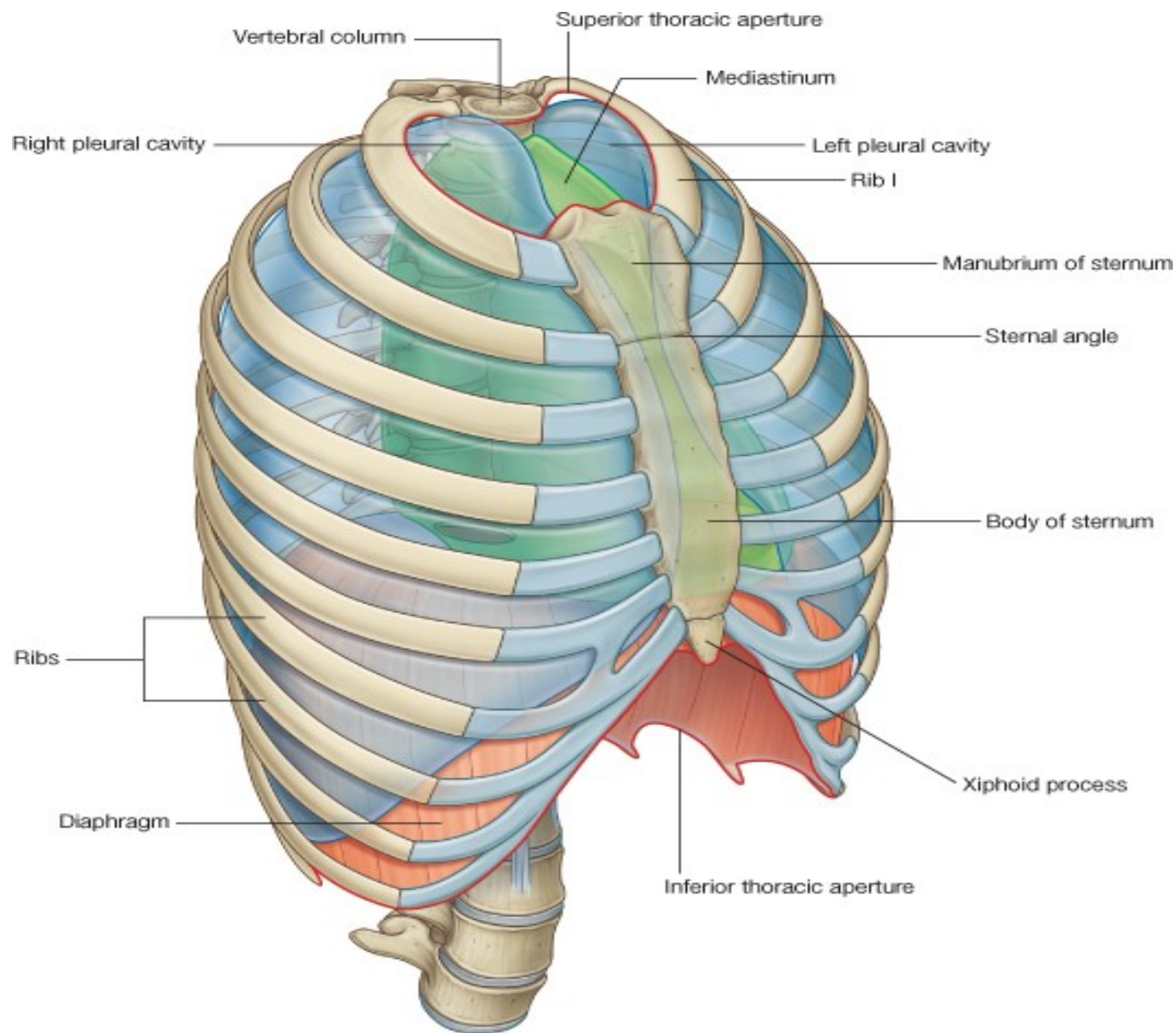
BONES OF THE THORAX:

- thoracic vertebrae
- Ribs
- Sternum

THORACIC VERTEBRAE:

- All thoracic vertebrae form joints with the ribs.
- The first thoracic vertebra contains a facet for first rib and upper part of the second rib.





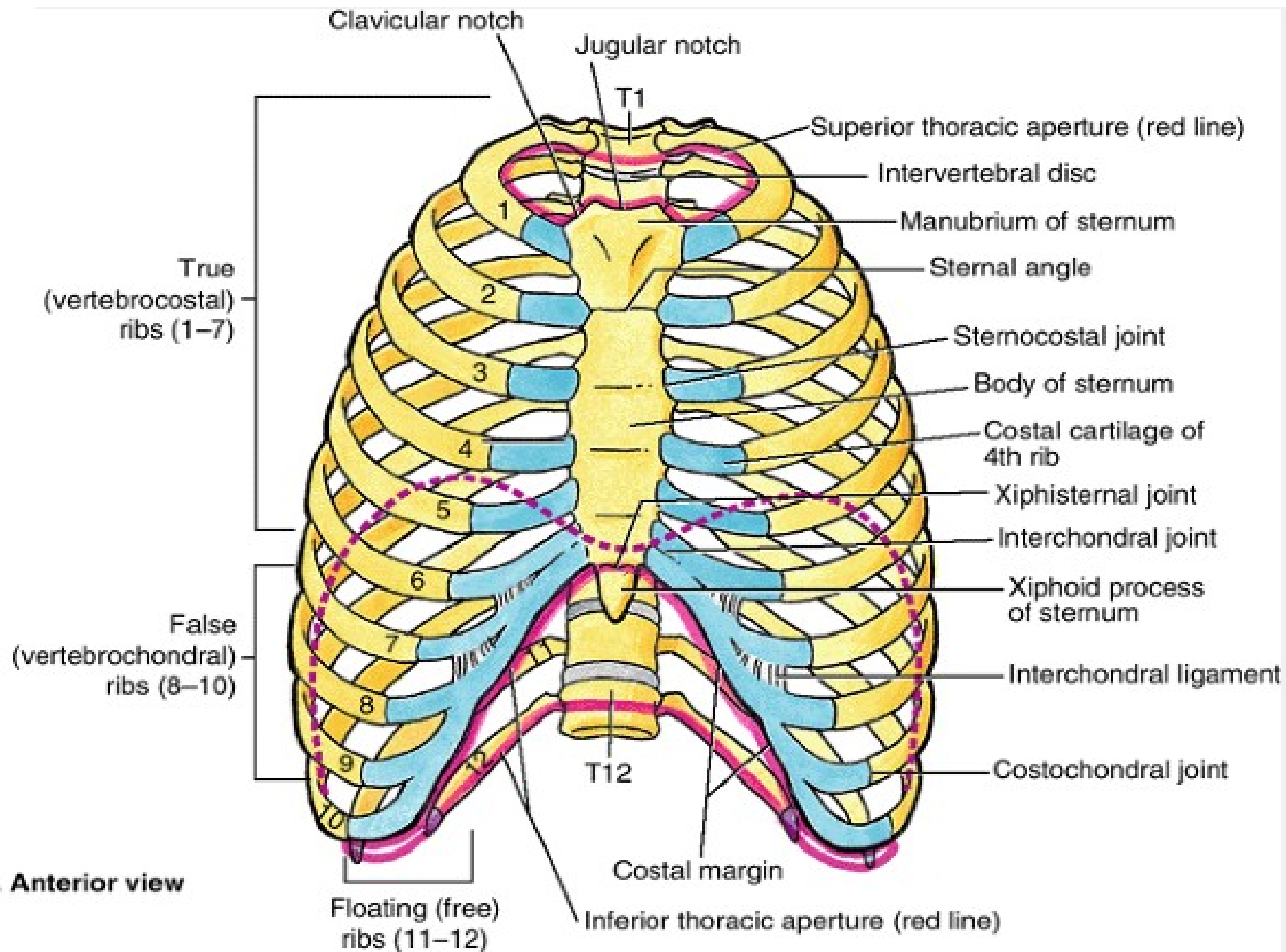
Skeleton of Thoracic Wall

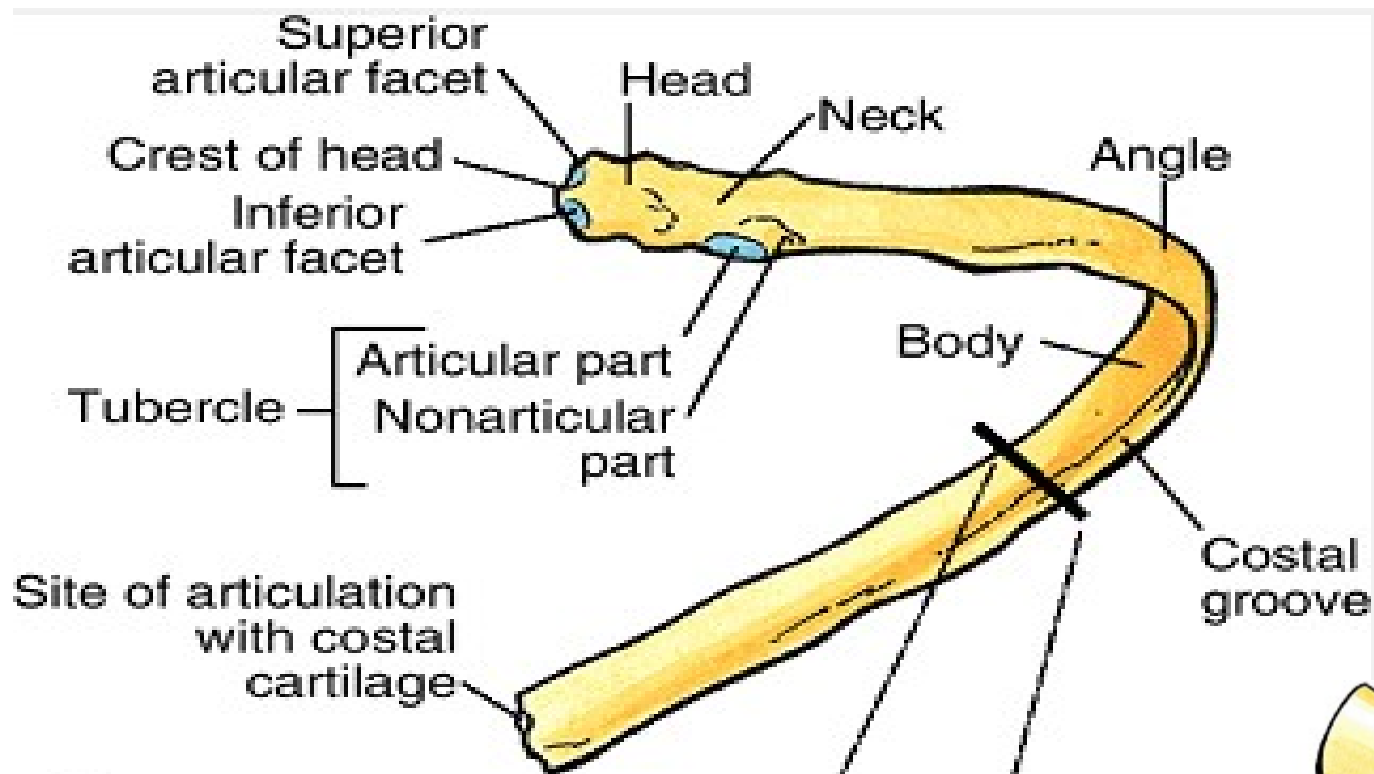
- thoracic skeleton forms osteocartilaginous thoracic cage .
- thoracic skeleton (bony thorax) includes;
 - 12 pairs of ribs and costal cartilages,
 - 12 thoracic vertebrae & intervertebral discs
 - sternum.
- Costal cartilages form the anterior continuation of the ribs, providing a flexible attachment at their articulation with the sternum .
- Ribs & their cartilages are separated by **intercostal spaces**, containing intercostal muscles, vessels, and nerves.

Skeleton of Thoracic Wall

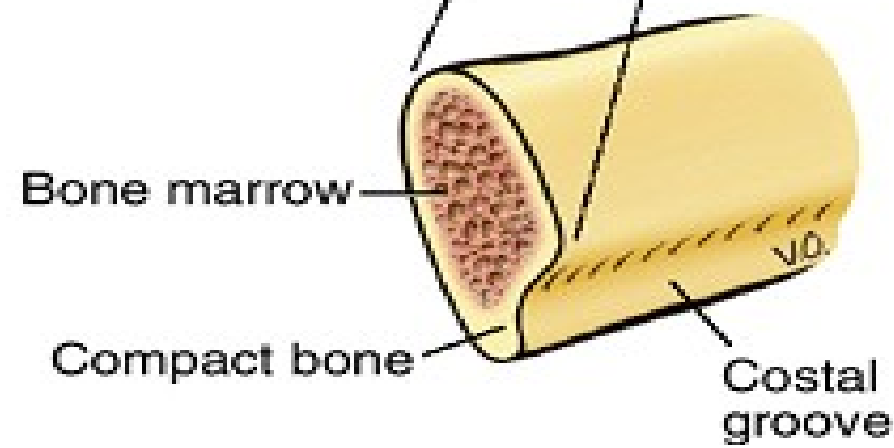
Ribs:

- The ribs are **curved, flat bones** that form most of the thoracic cage .
- They are remarkably light in weight yet highly resilient.
- There are three types of ribs:
 1. **True** (vertebrocostal) **ribs** (1st- 7th ribs) attach directly to the sternum through their own costal cartilages.
 2. **False** (vertebrochondral) **ribs** (8th -10th ribs) have cartilages that are joined to the rib just **superior** to them; → thus, their connection with the sternum is **indirect**.
 3. **Floating** (free) **ribs** (11th- 12th ribs) have rudimentary cartilages that do not connect even indirectly with the sternum; → instead, they end in the posterior abdominal musculature.

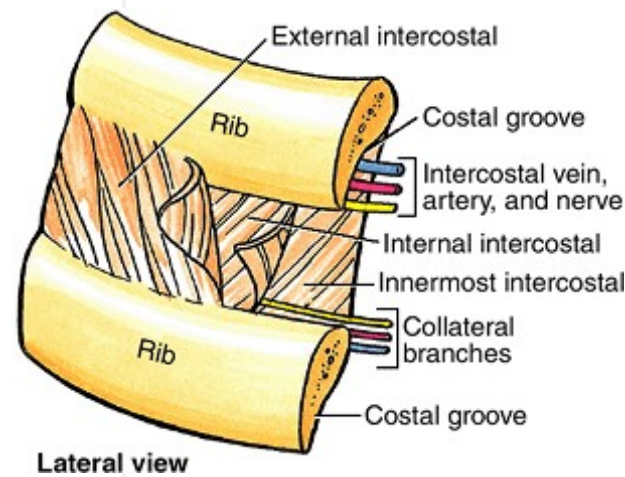




A Posterior view



B
Costal groove
Compact bone
Bone marrow



Lateral view

Skeleton of Thoracic Wall

Sternum:

- The sternum is the flat, vertically elongated bone that forms the middle of the anterior part of the thoracic cage.
- The sternum consists of three parts: **manubrium**, **body**, and **xiphoid process**

The xiphoid process:

- is an important landmark in the median plane because
 - Its junction with the sternal body at the xiphisternal joint indicates the inferior limit of the central part of the thoracic cavity projected onto the anterior body wall
 - this joint is also the site of the infrasternal angle (subcostal angle) of the inferior thoracic aperture
 - **It is a midline marker for the superior limit of:**
 - the liver
 - the central tendon of the diaphragm
 - the inferior border of the heart

Manubriosternal joint

- *can be palpated easily.*
- *referred to as the sternal angle(of Louis). This elevation marks the site of articulation of rib II with the sternum. Rib I is not palpable because it lies inferior to the clavicle and is embedded in tissues at the base of the neck. Therefore, rib II is used as a reference for counting ribs and can be felt immediately lateral to the sternal angle.*
- In addition, the **sternal angle** lies:
 - ✓ between vertebrae T4 and T5
 - ✓ separates the superior mediastinum from the inferior mediastinum and
 - ✓ marks the superior border of the pericardium.
 - ✓ passes through the end of the ascending aorta and the beginning of the arch of the aorta,
 - ✓ end of the arch of the aorta and the beginning of the thoracic aorta,
 - ✓ bi-furcation of the trachea, and just superior to the pulmonary trunk

Thorax

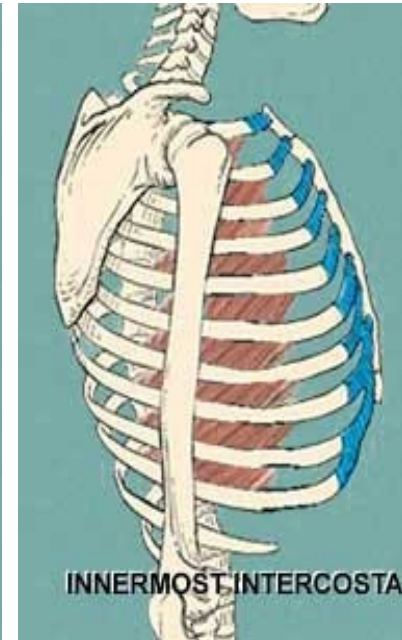
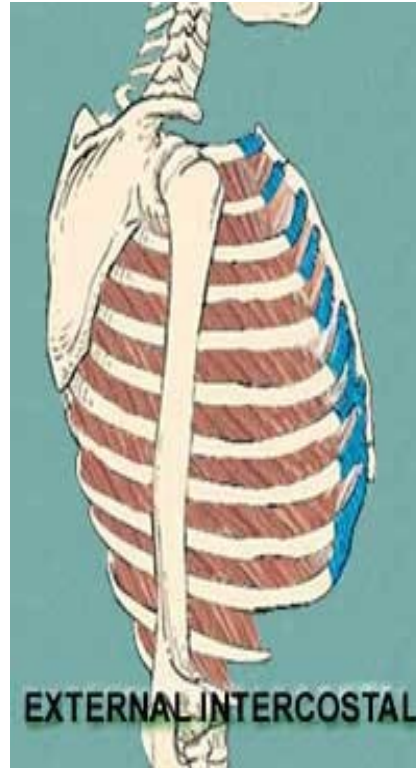
1. major thoracic organs of the thorax are:
 - **lungs**
 - **heart**
2. The other structures include:
 - ***aorta and its branches***
 - ***superior vena cava & inferior vena cava***
 - ***trachea and primary bronchi; esophagus***
 - ***Nerves: sympathetic trunks & their associations; vagus & phrenic nn***
 - ***azygos and hemiazygos venous systems***
 - ***Thoracic duct***

Muscles of the Thoracic Wall

- The three layers of the intercostal muscles are:
- ***external layer -- external intercostal***
- ***internal layer -- internal intercostal***
- ***innermost layer -- transversus thoracic (anterior), innermost (lateral) and subcostal (posterior).***
- The **diaphragm** is the most important muscle of the thoracic wall: ***primary component of normal/silent respiration.***
- ***innermost layer*** is split into ***transversus thoracis, innermost intercostal and subcostal muscles*** & these make up the deepest layer of muscles from anterior to posterior, respectively.

Muscles of the Thoracic Wall

- ***external intercostal***
- ***internal intercostal***
- ***innermost intercostal:***
- ***transversus thoracis (anterior)***
- ***Innermost (lateral)***
- ***subcostal (posteriorly)***



Muscles of Thoracic Wall: Diaphragm

Diaphragm:

- **Central Tendon:** is approximately in the center of the dome → Heart and pericardium rest on it._
- The position of the diaphragm is influenced by several factors. The important ones are the 1. elastic retraction force of the lungs 2. the pressure in the abdominal cavity and 3. the gravitation force of the underlying viscera. When a person is sitting or standing the diaphragm operates at its lowest level.

Openings of the Diaphragm: has 3 large openings:

1. Opening for inferior vena cava:

- *at level of T8 vertebra; in central tendon part*
- *Inferior vena cava & Rt. Phrenic n. transmitted.*

2. Esophageal hiatus:

- *at T10 vertebra; in muscular part*
- *Esophagus, Rt. & Lt. vagus nerves, Lt Gastric vessels (esophageal branches) & lymphatic vessels transmitted.*

3. Aortic hiatus:

- *At level of T12 vertebra; in muscular part: b/n the crura*
- *Aorta, Thoracic duct & Azygos vein transmitted*

Respiration

- During **restful breathing**, the *diaphragm* does most of the work.
- During **increased need for oxygen** (exercise, pathology), the *lateral and anteroposterior* movements will be increased: by action of *intercostal muscles*.
- When the thoracic muscles can no longer do the job, other muscles attaching to the thorax will be called into action (*pectoralis major and minor, sternomastoid*, etc.)
- During expiration, the *intercostal muscles & diaphragm relax* and the elastic fibers of the *lung* and the *costal cartilages recoil* to their original state before inspiration.
- The automatic nature of the respiratory cycle is controlled in the *respiratory centers of the brain stem*.

The Lungs

The Lungs:

- The lungs are the essential organs of respiration.
- They are highly elastic structures.

✂️➔ It is this elasticity which is responsible for the ordinary expiration force.

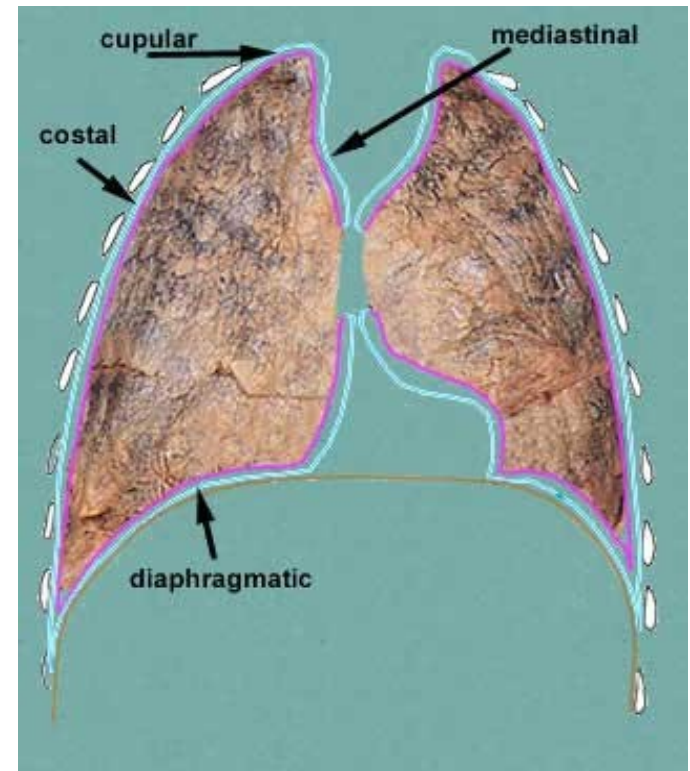
- Each lung lies free in its own pleural cavity.

✂️➔ Its only attachment is at its root.

✂️➔ By “root of the lung” we mean all structures entering and leaving the lung at the hilus. ➔ They connect the lung with trachea (by Bronchii) and heart (by pulmonary vessels).

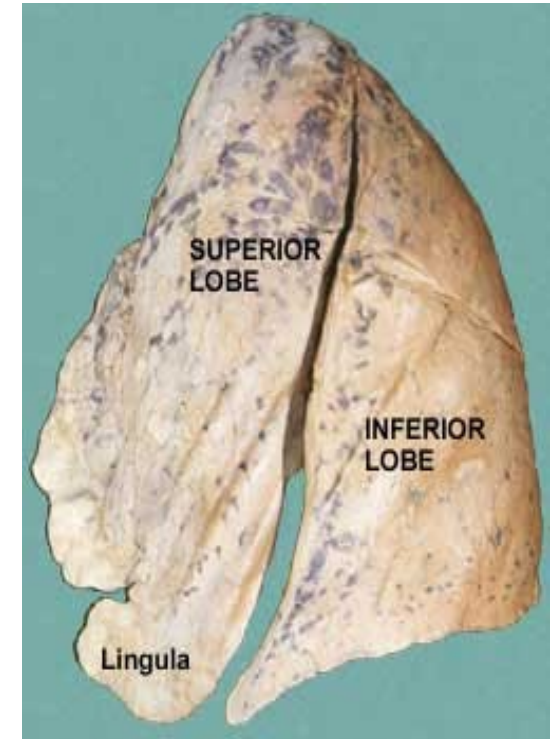
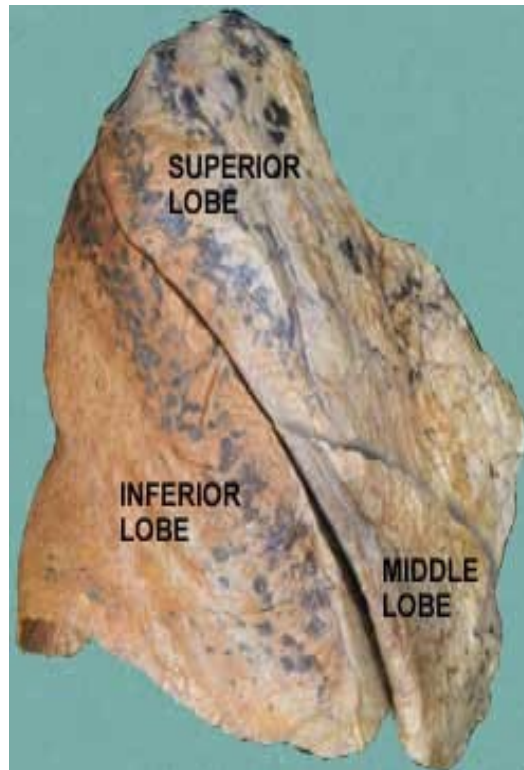
The Lungs

- The root of the lung lies opposite T5, T6, T7.
- The Root/hilum is covered by a sleeve of parietal pleura.
- The lungs present an apex, a base, 3 borders and 2 surfaces:
- The apex projects above the thoracic inlet into the root of the neck.
✂ ➔ It is related to subclavian, common carotid, brachiocephalic arteries, brachiocephalic veins, scalenus medius, sympathetic trunk, 1st thoracic spinal nerve and superior intercostal arteries.
- The base or diaphragmatic surface is related to the diaphragm.

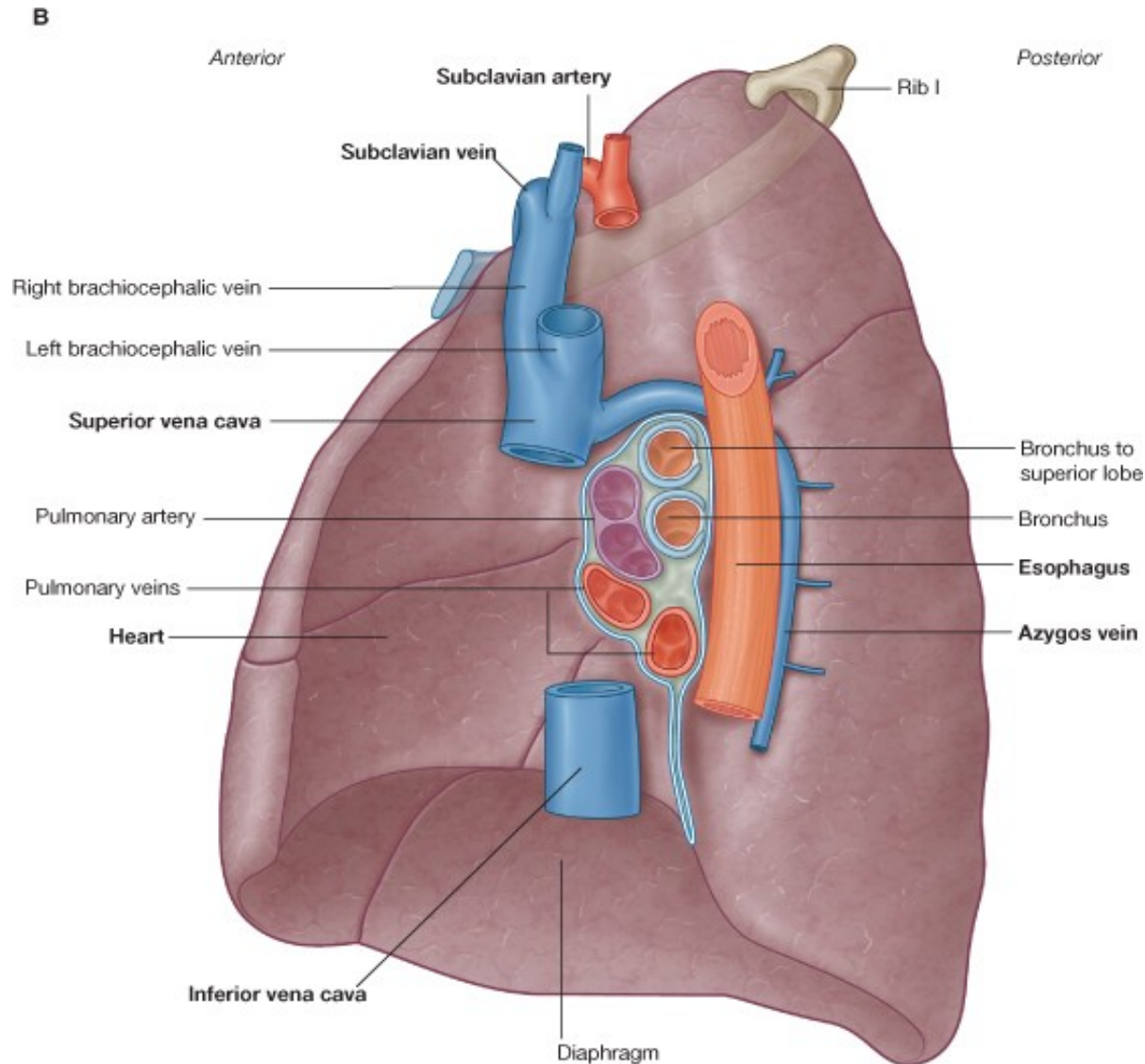


Lobes & Surface of the Lungs

- The lungs fill lateral thoracic cavities & are divided into lobes:
→ **Left lung has 2 lobes & Right lung has 3 lobes.**
- bulk of the lung surface is against the ribs & is called **costal surface.**
- Other surfaces include the:
diaphragmatic & mediastinal.

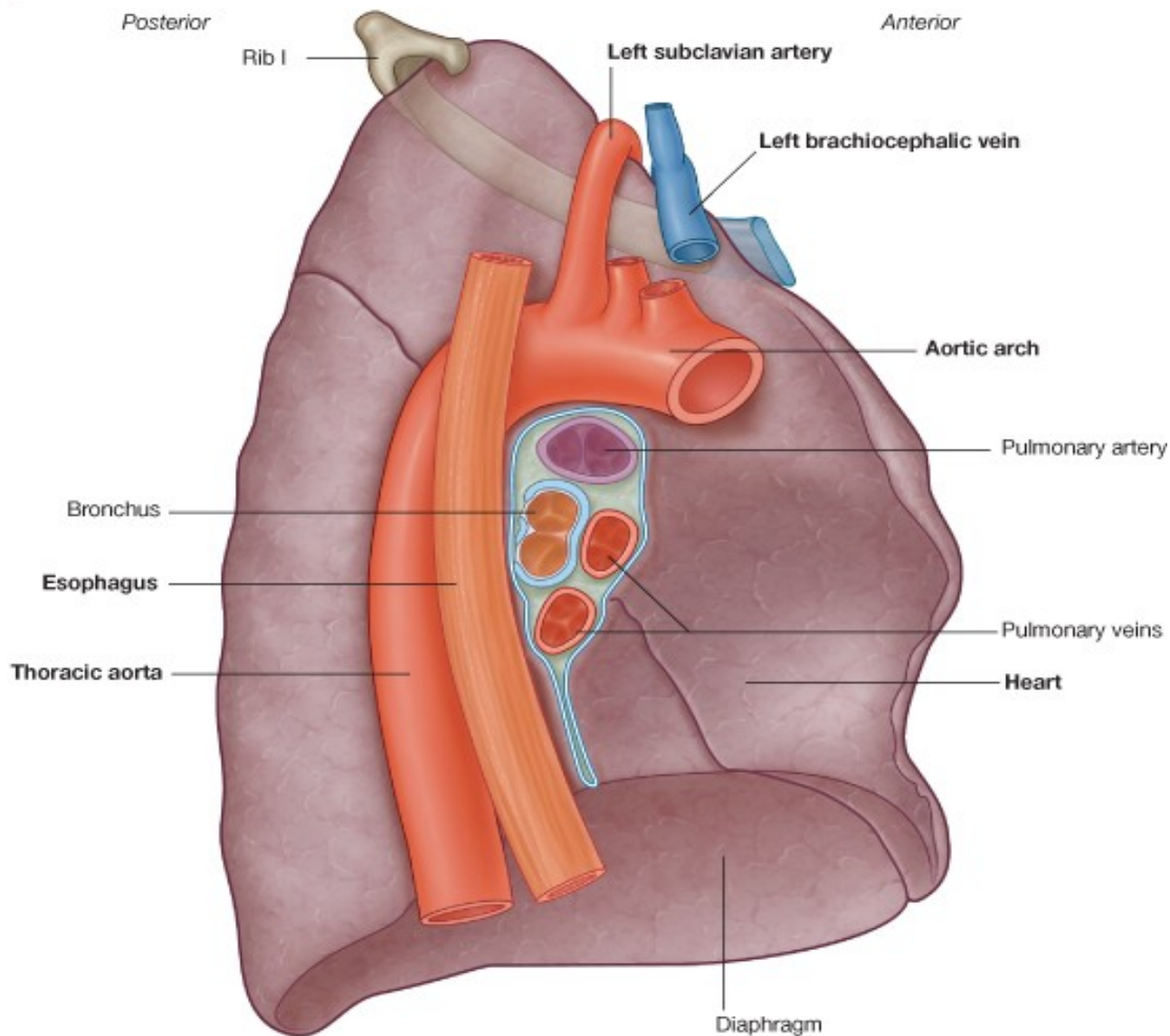


Root of the Right Lung



Root of the Left Lung

B



The Trachea & Bronchial Tree

The Trachea: commences in the neck at lower border of cricoid cartilage at the level of the 6th cervical vertebra, and ends below by dividing into a right and left primary bronchus at the level of the sternal angle (b/n 4th & 5th thoracic vertebral).

- The trachea is held open by 'C-shaped' transverse cartilage rings embedded in its wall-

✂️ ➔ open part of the 'C' faces posteriorly.

✂️ ➔ The posterior wall of the trachea is composed mainly of smooth muscle.

- It lies in the midline except at its bifurcation where it slightly deviates to the right.

Trachea



Right primary bronchus

Left primary bronchus



3 Lobar bronchi



2 Lobar bronchi



10 segmental bronchi



10 segmental bronchi

Bronchioles



Terminal Bronchioles



Respiratory Bronchioles



Alveolar ducts



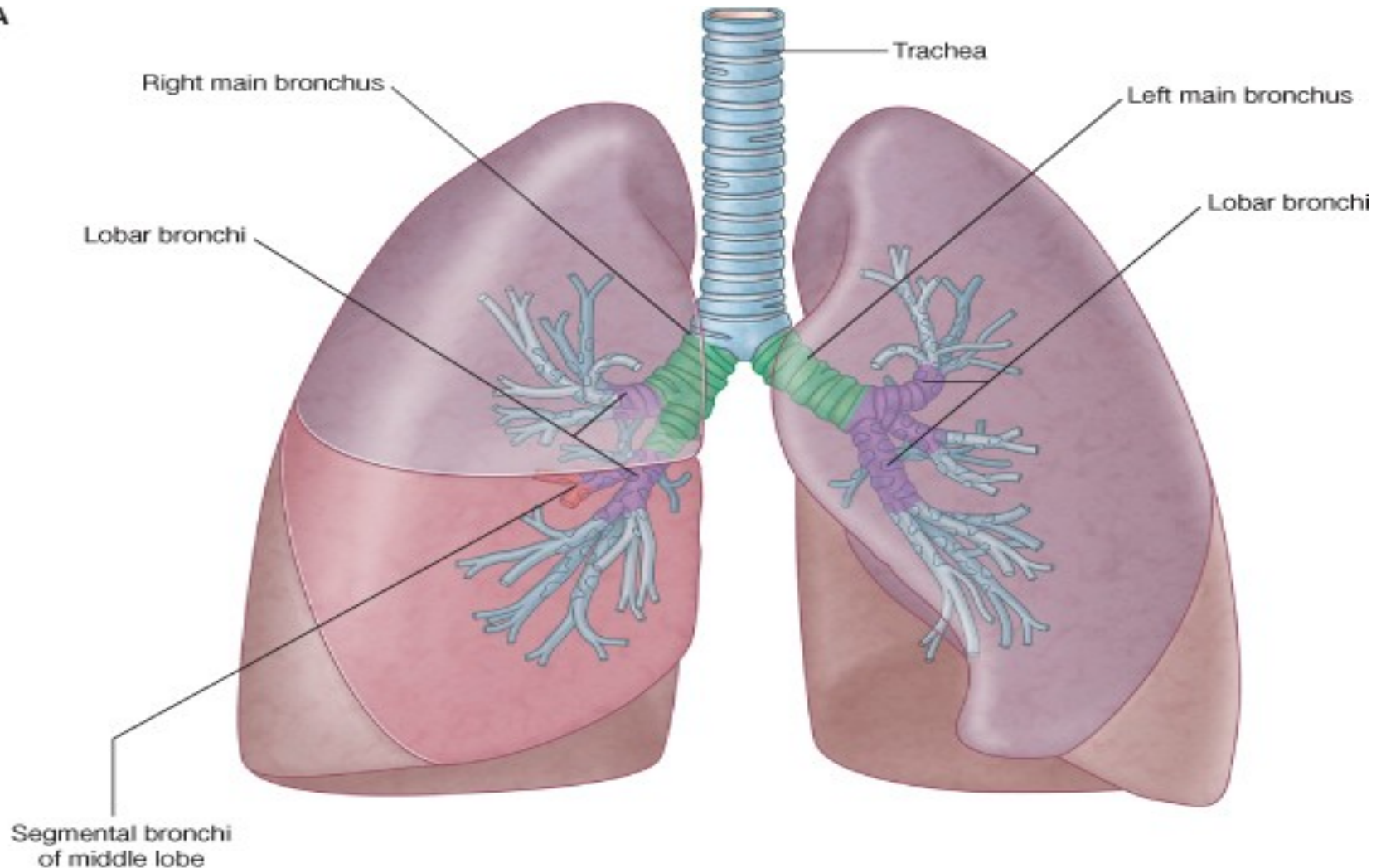
Alveolar sac



Alveolus

The Trachea & Bronchial Tree

A



© Elsevier. Drake et al: Gray's Anatomy for Students - www.studentconsult.com

The Trachea (Cervical Portion)

- Whole length = 12 cm
- Lower level of **cricoid cartilage(C6)** to **level of T₅**
- Descends in the midline of the neck as a continuation of the larynx

Relations

□ Anteriorly –

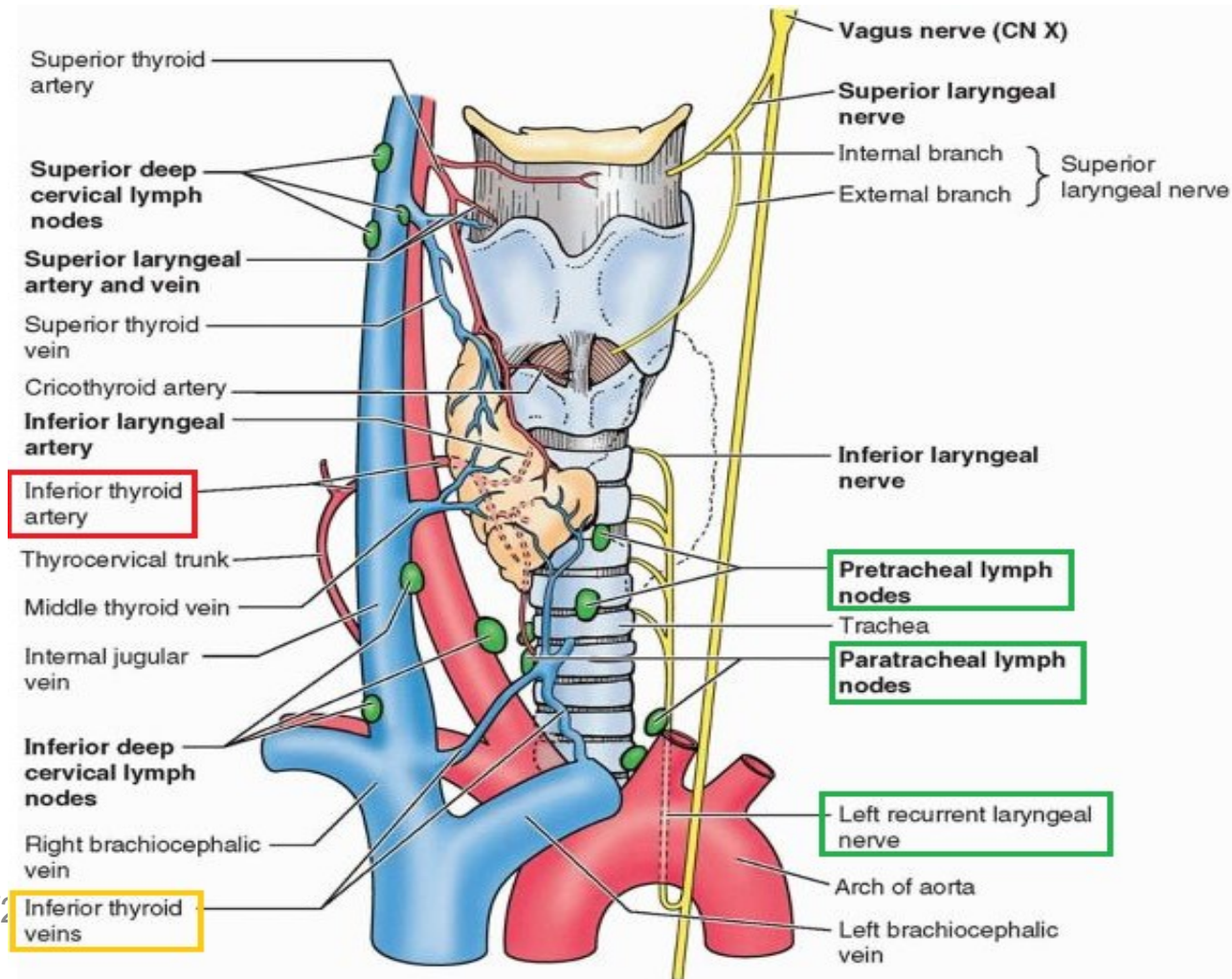
- ✓ Isthmus of thyroid gland in front of 2nd, 3rd, 4th rings
- ✓ Below isthmus – inferior thyroid vein, thyroid ima (if present)
- ✓ At the root of the neck – left innominate vein & jugular arch

□ Laterally

- ✓ Lobe of thyroid gland
- ✓ Carotid sheath and its content

□ Posteriorly

- ✓ Esophagus
- ✓ Recurrent laryngeal nerve
- ✓ Vertebral column



Arterial supply: branch from inferior thyroid artery

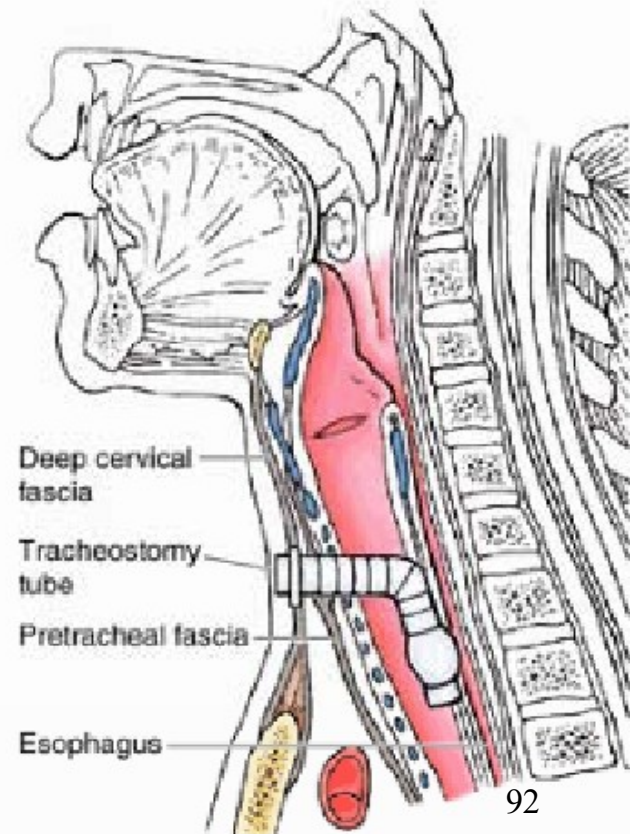
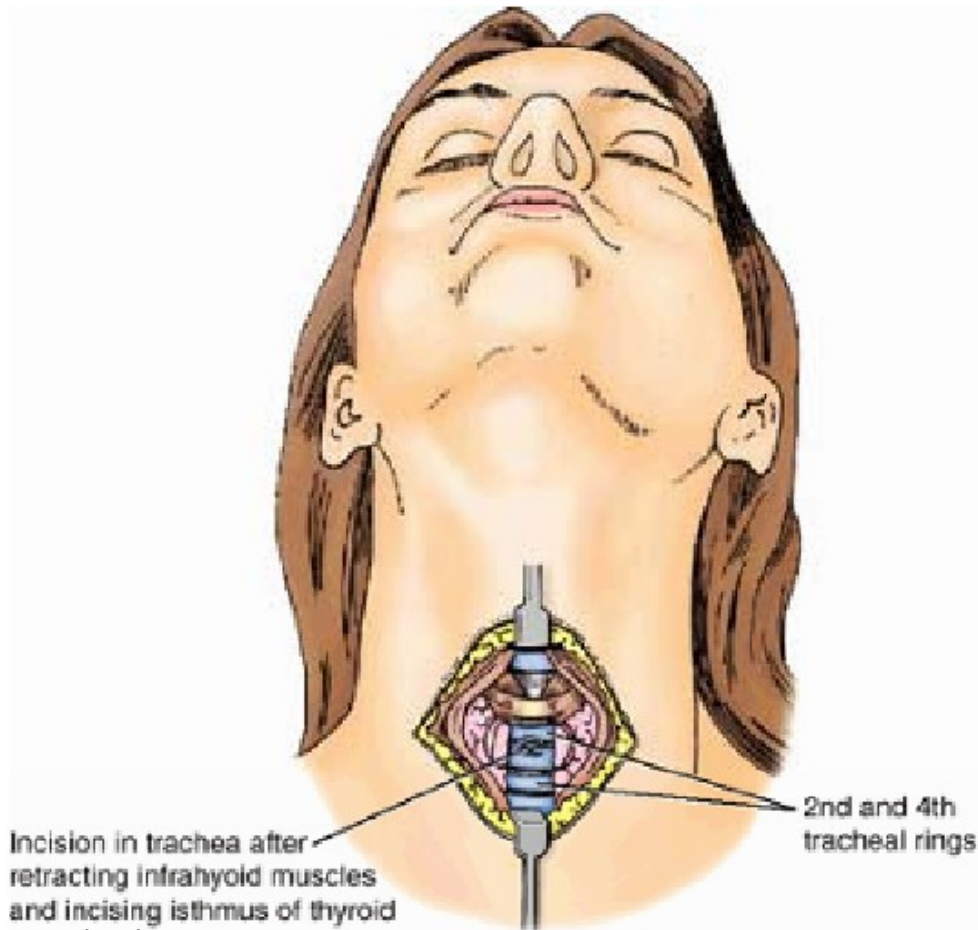
Venous drainage: inferior thyroid vein

Lymphatic drainage: pretracheal and paratracheal lymph node

Nerve supply – branches from vagi, recurrent laryngeal &
sympathetic

Clinical note:

- tracheostomy surgery is usually performed inferior to the thyroid isthmus;



Esophagus (cervical portion)

- Whole length = 25 cm with 3 parts; cervical, thoracic & abdominal
- **Begins:** as a continuation of pharynx at the level of cricoid cartilage
- **Ends:** in the abdomen opposite to T₁₁ by joining the cardiac orifice of the stomach
- **Course** – begin in the median plane then turns slightly to the left as it descends in the neck

Relations

Anteriorly: trachea and recurrent laryngeal N

Posteriorly – longus coli and vertebral column

Laterally – lobe of thyroid gland, carotid sheath & its content at the root of the neck

- **To the right side:** cervical pleura
- **To the left side:** subclavian artery and thoracic duct

Arterial supply: branches from inferior thyroid A

Venous drainage: inferior thyroid vein

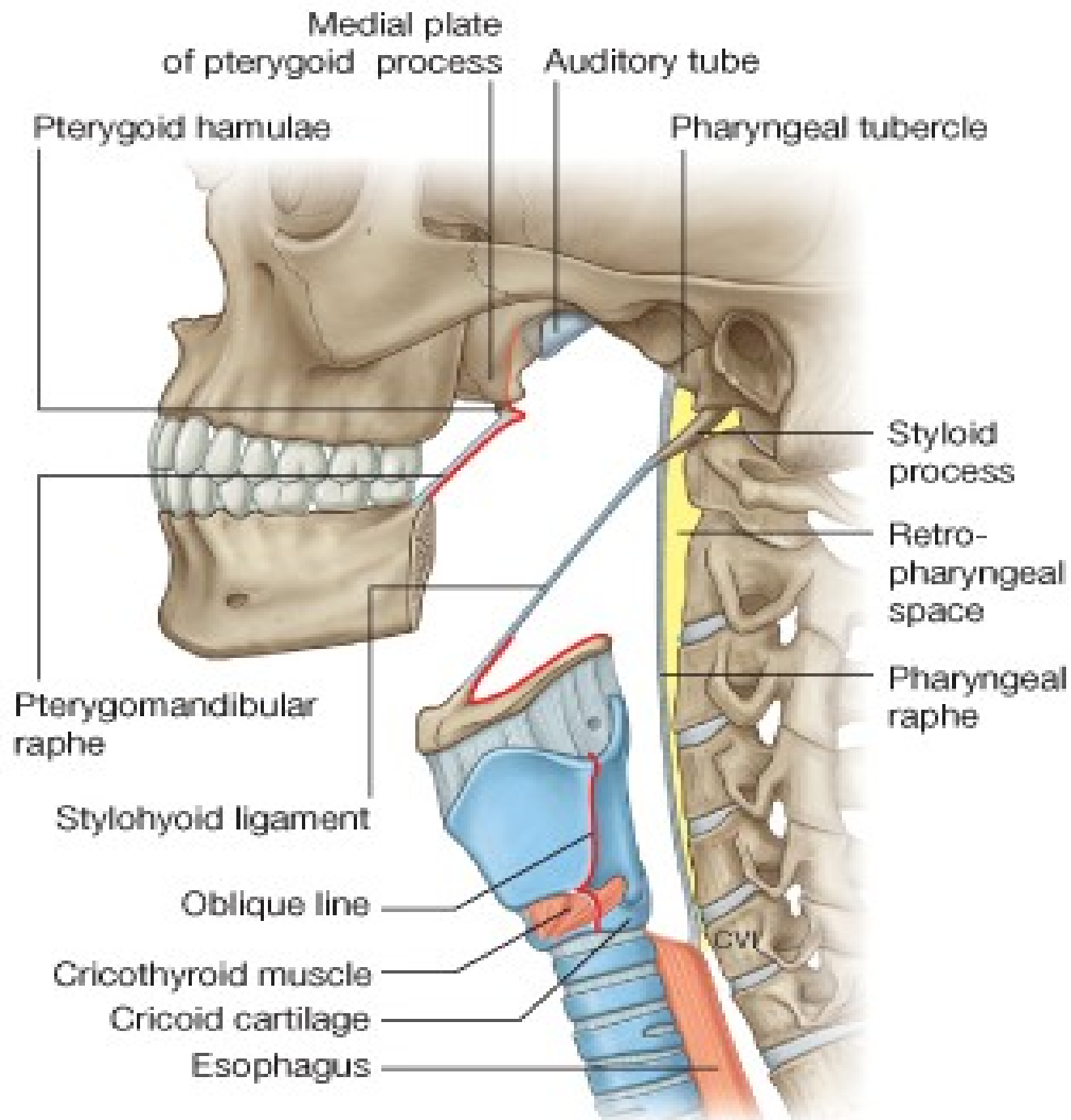
Lymphatic drainage: deep cervical lymph nodes

Nerve supply: recurrent laryngeal & sympathetic trunk

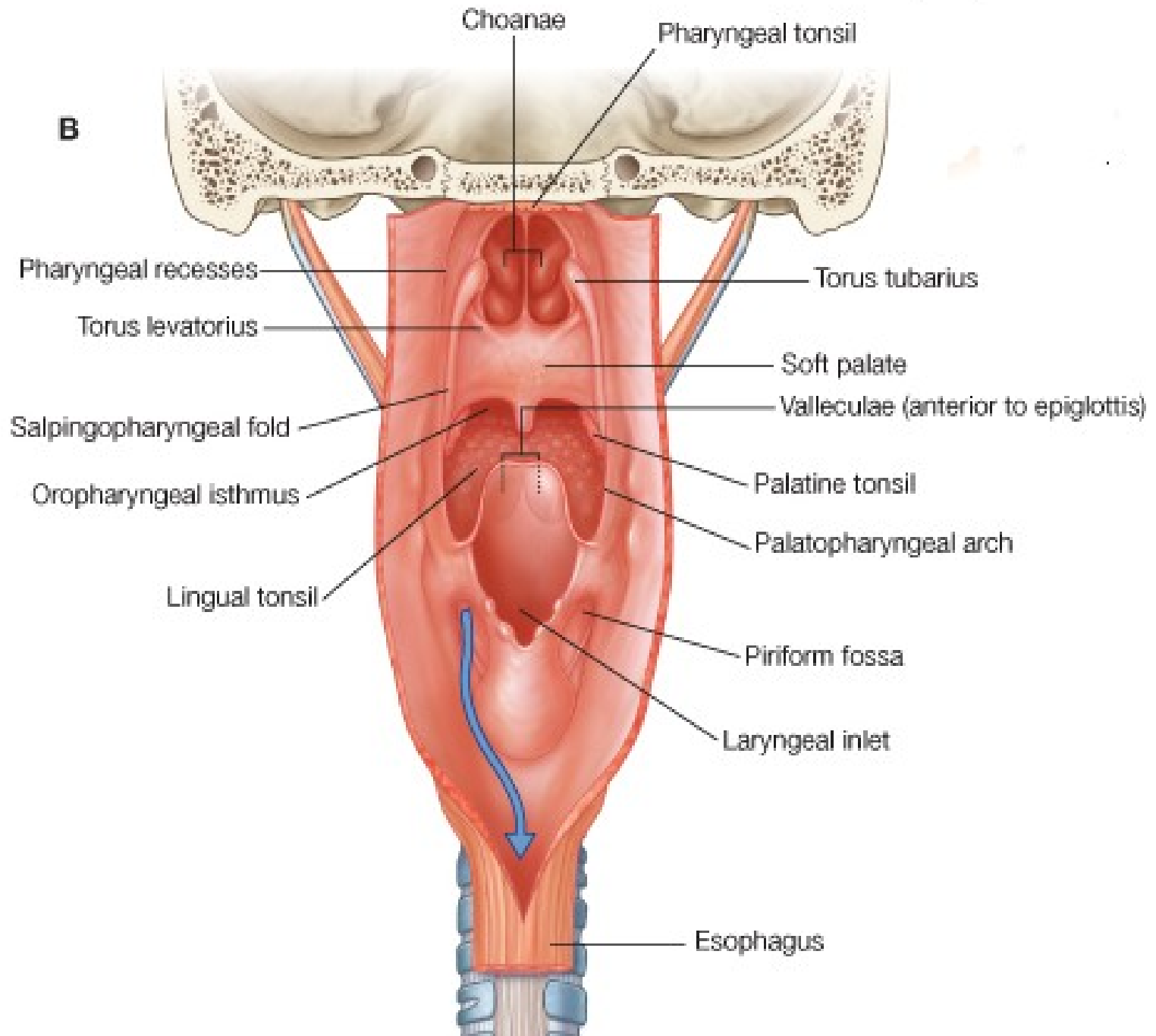
Pharynx

- Funnel shaped **fibromuscular tube** found behind nasal, oral and laryngeal cavities
- Extends from the base of skull to inferior border of cricoid cartilage (C6) = 12 cm long
- Becomes continuous with the esophagus at C6

- Boundaries and relation
 - **Superior** – body of sphenoid and basilar part of occipital
 - **Inferior** – continuous with esophagus
 - **Posterior** – prevertebral fascia
 - **Anterior** – communicate with nasal cavity, oral cavity and larynx
 - **Lateral** – attached to medial pterygoid plate, pterygomandibular raphe, mandible, tongue, hyoid, thyroid and cricoid cartilage
- 3 regions: nasal (nasopharynx), oral (oropharynx) and laryngeal (laryngopharynx)



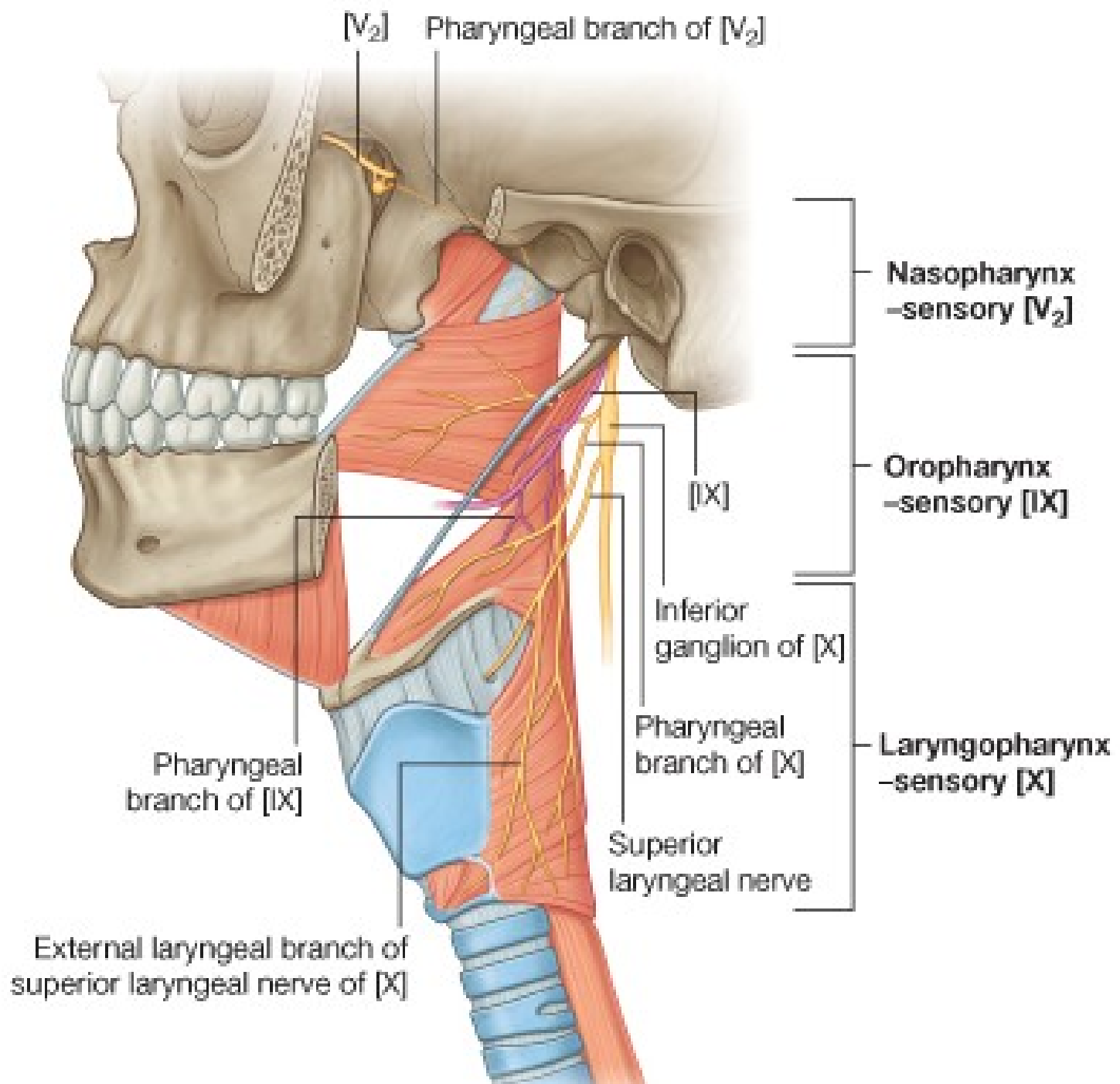
- **Walls of the pharynx**
 - Composed of (inside out)
 - ✓ Mucus membrane
 - ✓ Submucosa
 - ✓ Pharyngobasilar fascia
 - ✓ Muscles
 - ✓ Bucopharyngeal fascia



- **Pharyngobasilar fascia**
 - Covers the internal surface of the constrictor muscles
 - Connects pharynx with the base of the skull and fill the gap between the superior constrictor and skull
 - Together with the buccopharyngeal fascia fill the gaps in the muscular wall
- **Buccopharyngeal fascia**
 - covers the external surface of the buccinator muscle and the superior constrictor muscle
 - contains: pharyngeal venous plexus and pharyngeal plexus of nerves

- **The pharyngeal plexus of veins**
 - lies deep to this fascia, draining the pharynx, soft palate and the pharyngeal tonsil
 - communicates with the pterygoid plexus of veins and drains into the internal jugular vein
 - These veins have no valves and infection from the pharynx, palate and tonsils may spread into the meningeal venous plexus

- **The pharyngeal nervous plexus**
 - formed by branches from CN IX, X and superior cervical ganglion
 - **Motor branches** – CN X (cranial root of accessory nerve)
 - supply all muscles of pharynx and palate except stylopharyngeus (CN IX) and tensor palati (CN V3)
 - inferior constrictor receive additional fibers from external and recurrent laryngeal nerves
 - **Sensory branches**
 - nasopharynx - maxillary nerve (CN V2)
 - oropharynx - CN IX
 - laryngopharynx - vagus

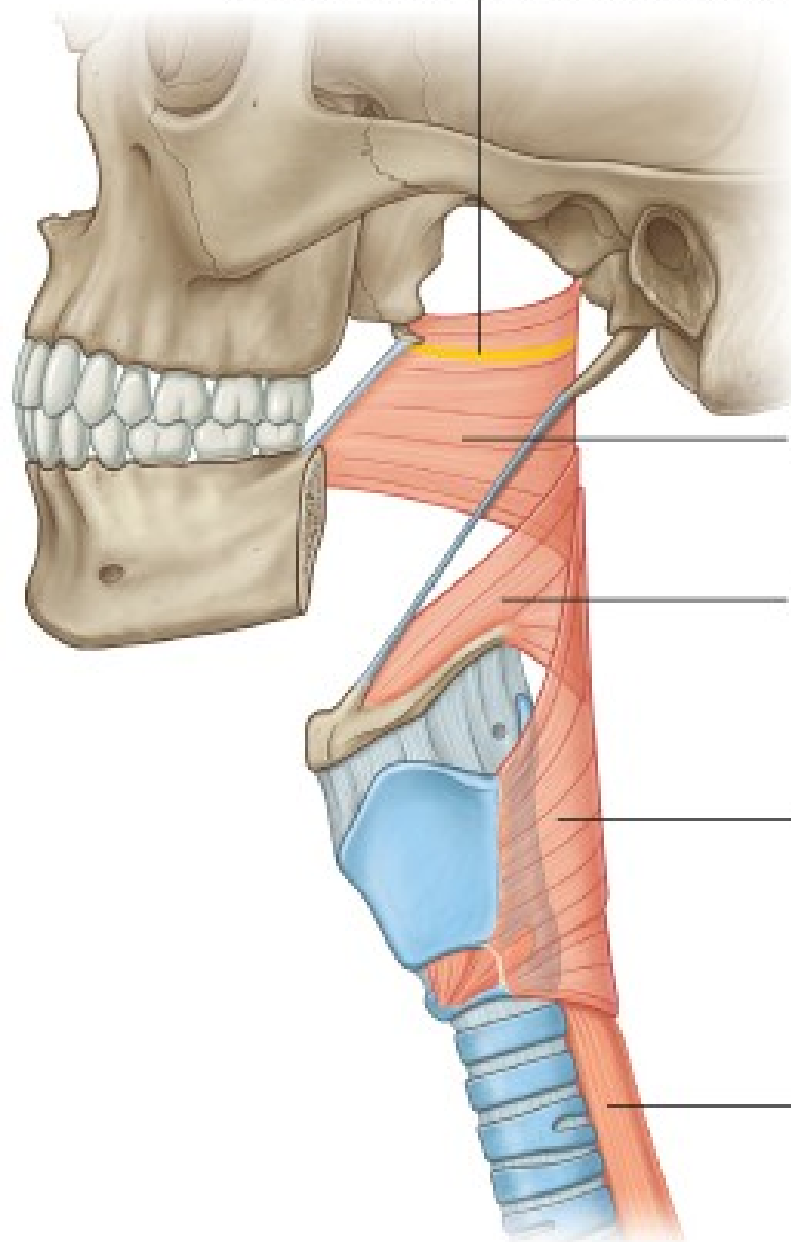


- **Muscular layer**

- Inner longitudinal and outer circular skeletal muscles
- 6 paired skeletal muscles
 - **Three pairs of circular muscles** (external muscles): superior, middle and inferior constrictor muscles
 - **Three pairs of longitudinal** (internal muscles): palatopharyngeus, stylopharyngeus & salpingopharyngeus

A

Position of palatopharyngeal sphincter
on deep surface of superior constrictor

**B**

Pharyngeal fascia

Pharyngeal
tubercle

Styloid
process

Stylohyoid
ligament

Stylopharyngeus
muscle

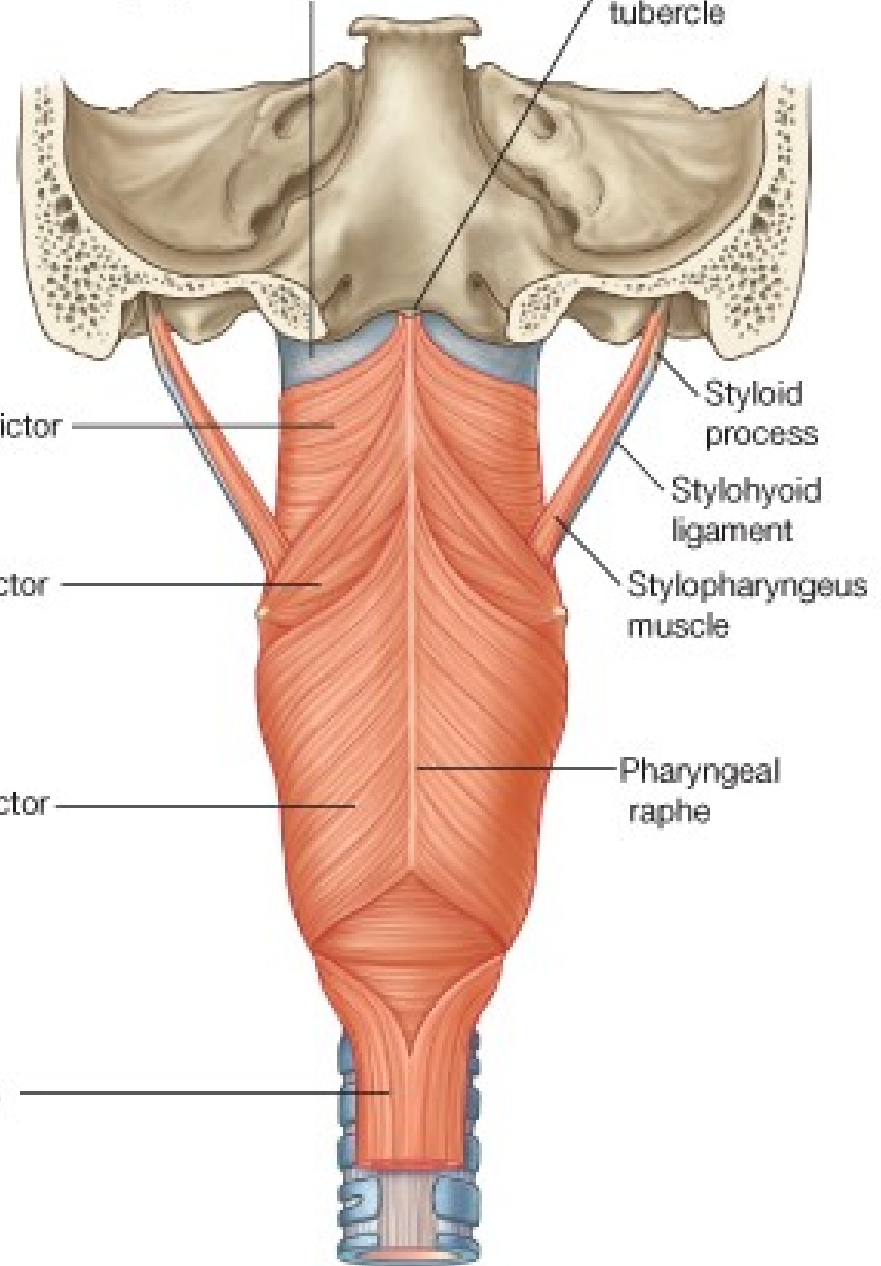
Pharyngeal
raphe

Superior constrictor

Middle constrictor

Inferior constrictor

Esophagus



- **Superior constrictor muscle**
 - lies behind the nasal and oral cavities and continuous with the plane of the buccinator
 - **Origin** – pterygoid hamulus, pterygomandibular raphé, posterior end of mylohyoid line of mandible & the side of the tongue
 - **Insertion** – median raphé of pharynx and pharyngeal tubercle

- **Middle constrictor muscle**

- lies behind the tongue and larynx
- It is truly a constrictor and plays a major role in swallowing
- **Origin** - stylohyoid ligament , greater and lesser horns of hyoid bone
- **Insertion** - median raphé of pharynx

- **Inferior constrictor muscle**

- **Origin** – oblique line of thyroid cartilage and the sides of cricoid cartilage
- **Insertion** – median raphé of pharynx
- The most inferior aspect is the cricopharyngeus which is the sphincter between the pharynx and the esophagus

- **Internal muscles of the pharynx**
 - Elevates larynx and pharynx during swallowing
- **Stylopharyngeus**
 - **Origin** - styloid process of temporal bone
 - **Insertion** - posterior & superior border of the thyroid cartilage
 - **Course** - descends between external & internal carotid arteries and enters the pharyngeal cavity with IX between the superior and middle constrictor muscles
 - Nerve supply - CN IX

- **Palatopharyngeus**

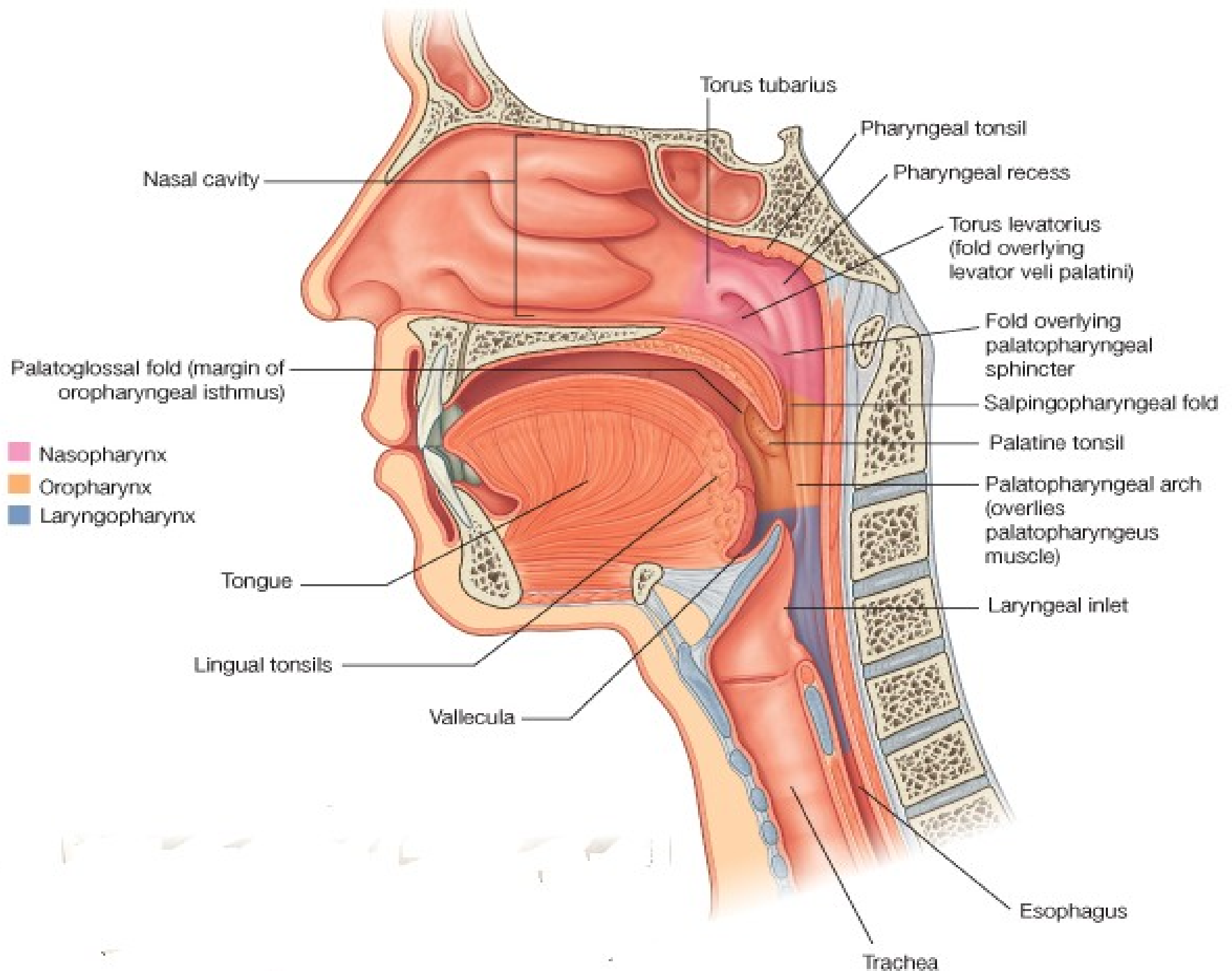
- Origin - hard palate and palatine aponeurosis
- Insertion – posterior border of lamina of thyroid cartilage and sides of pharynx
- Nerve supply – pharyngeal plexus
- The overlying mucosa forms the palatopharyngeal arch, posterior to the palatine tonsil

- **Salpingopharyngeus**

- Origin – cartilaginous part of auditory tube
- Insertion – blend with palatopharyngeus
- Nerve supply - pharyngeal plexus

- **Cavity of the pharynx**
 - 3 parts - Nasopharynx, Oropharynx & Laryngopharynx
- **Nasopharynx**
 - Behind nose and above lower border of soft palate
 - Respiratory
 - Walls
 - Roof – sphenoid & occipital bone, mucus membrane, submucosa (contain pharyngeal tonsil)
 - Floor – soft palate
 - Anterior wall – two posterior openings of the nasal cavity
 - Posterior wall - anterior part of atlas lined with mucus membrane

- Lateral wall – presents
 - Pharyngeal openings of auditory tube – behind inferior nasal concha
 - Tubal elevation - posterior to the auditory openings
 - Salpingopharyngeus fold – inferior to tubal elevation due to the muscle
 - Pharyngeal recess – depression behind tubal elevation
 - Tubal tonsil – lymphoid tissue in the submucosa behind the openings of the auditory tube
- Inferiorly – Nasopharynx communicate with oropharynx at nasopharyngeal isthmus



- **Oropharynx**

- Behind oral cavity, extends from soft palate to the upper border of epiglottis

- Walls

- Anterior

- Communicate with oral cavity through oropharyngeal isthmus (isthmus of fauces)

- » Laterally bounded by two arches – palatoglossal & palatopharyngeal; between them triangular fossa for palatine tonsil

- » Inferiorly pharyngeal part of tongue – contain lingual tonsil, 3 folds of mucus membrane

- Posterior wall – body of axis and upper part of C3 lined with mucus membrane

- **Laryngopharynx**

- posterior to the larynx, from superior border of epiglottis to inferior border of cricoid
- Walls
- Anterior
 - Communicate with larynx through inlet of larynx; bounded by epiglottis, aryepiglottic fold & inter arytenoid folds
 - Mucus membrane covering posterior surface of larynx
- Posterior – related to C4 - C6 vertebrae
- Lateral – mucus membrane covering the thyroid cartilage and thyrohyoid membrane presents piriform recess

- **Nerve supply to pharynx**
 - Mucus membrane is supplied with CN IX through pharyngeal plexus except
 - Most of mucus membrane of nasopharynx which is supplied by maxillary nerve
 - Mucus membrane of the anterior wall of laryngopharynx & around the inlet of the larynx which are supplied by internal laryngeal nerve from vagus

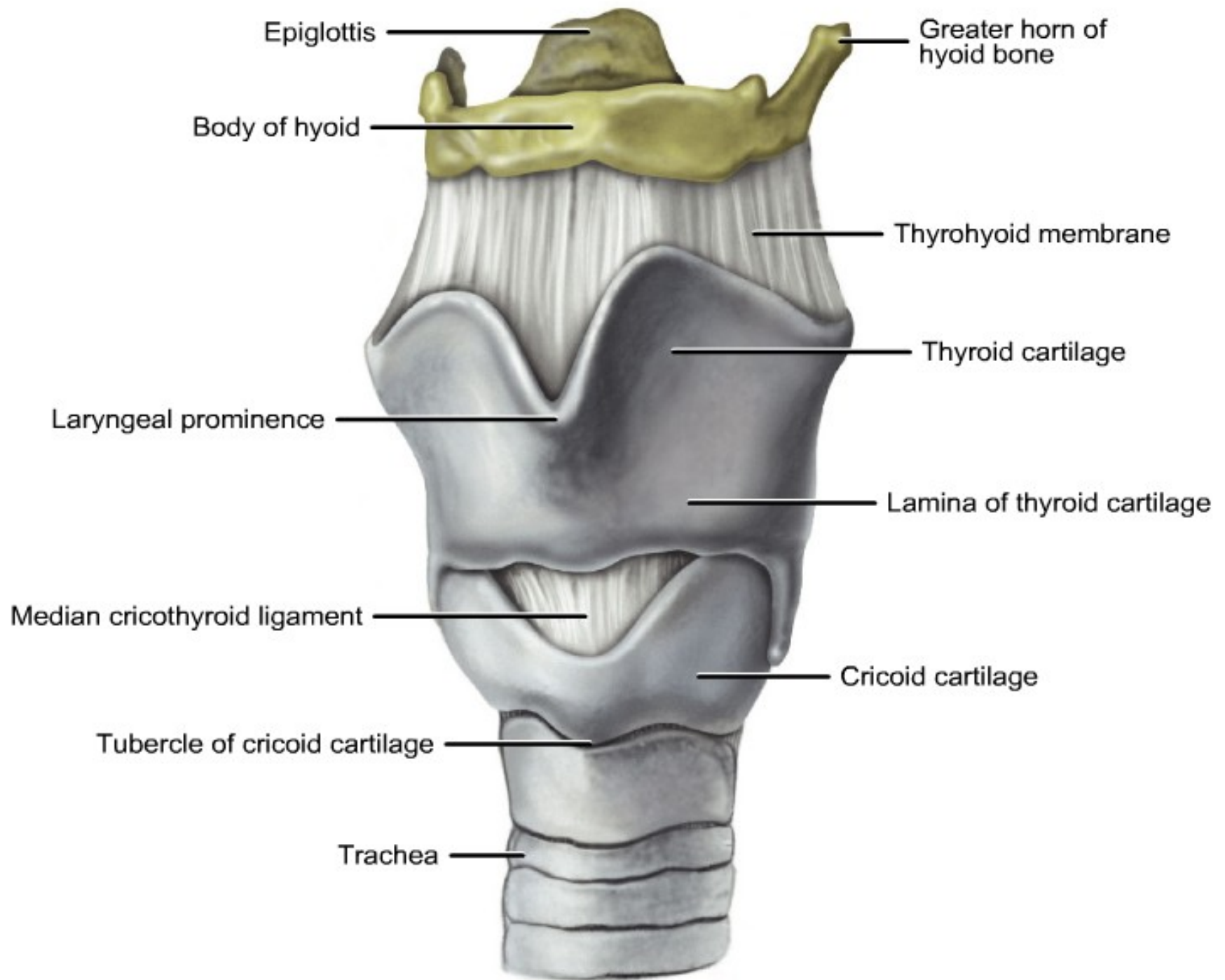
- **Arteries**
 - Ascending pharyngeal
 - Ascending palatine
 - Tonsilar branch of facial artery
 - Branches from lingual & maxillary
- **Veins** – end in pharyngeal plexus that ends in the internal jugular vein and facial vein
- **Lymphatic vessels** – end in the retropharyngeal lymph node that drain into the deep cervical lymph nodes

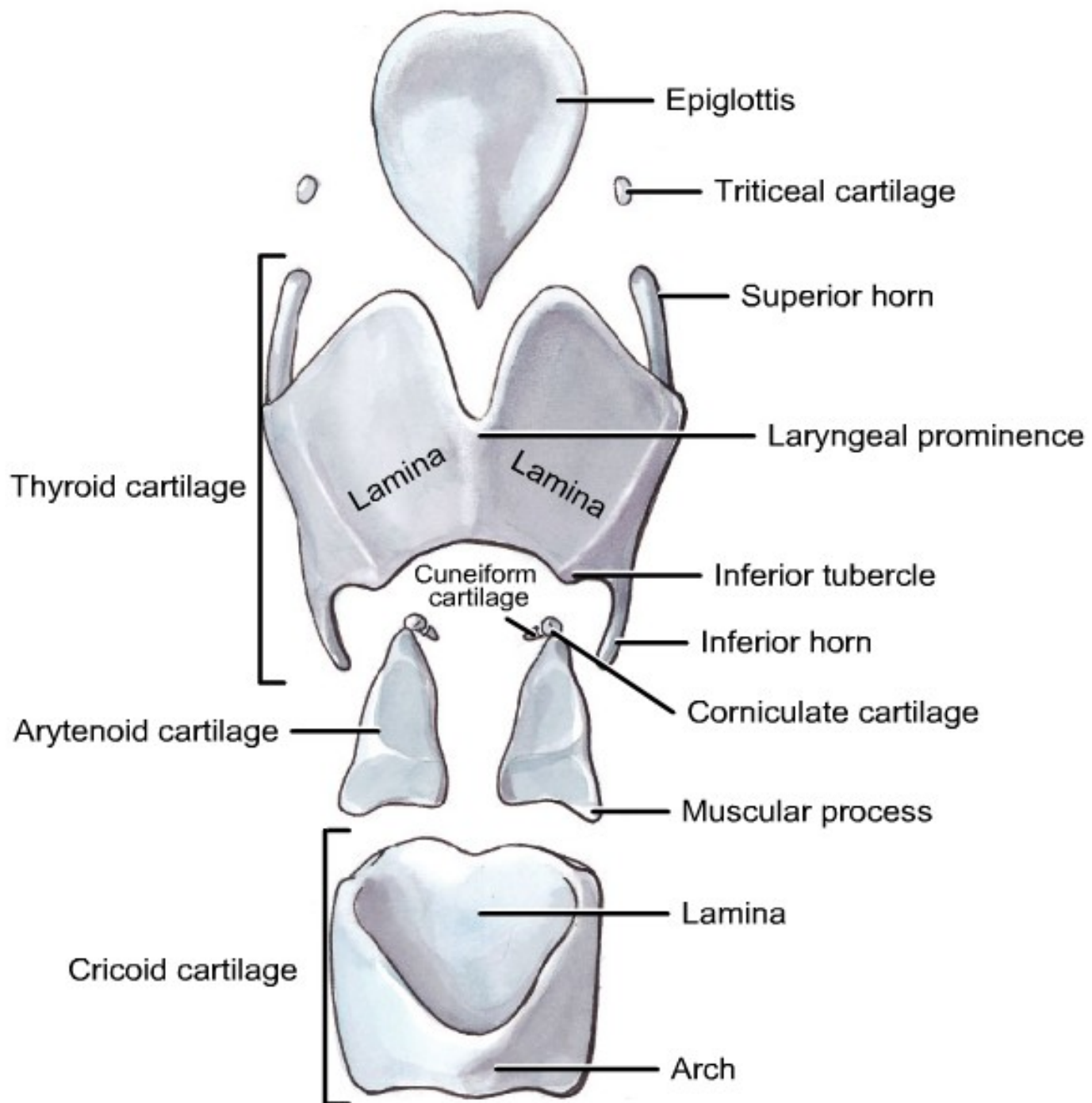
Larynx

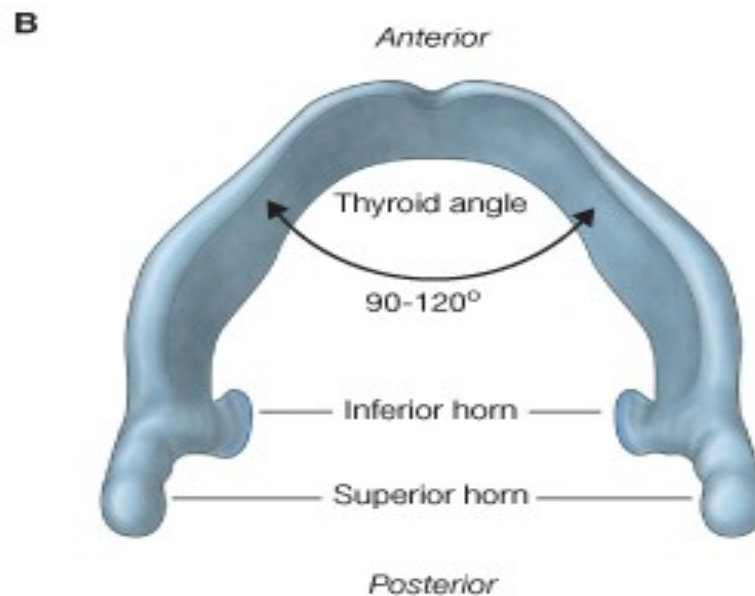
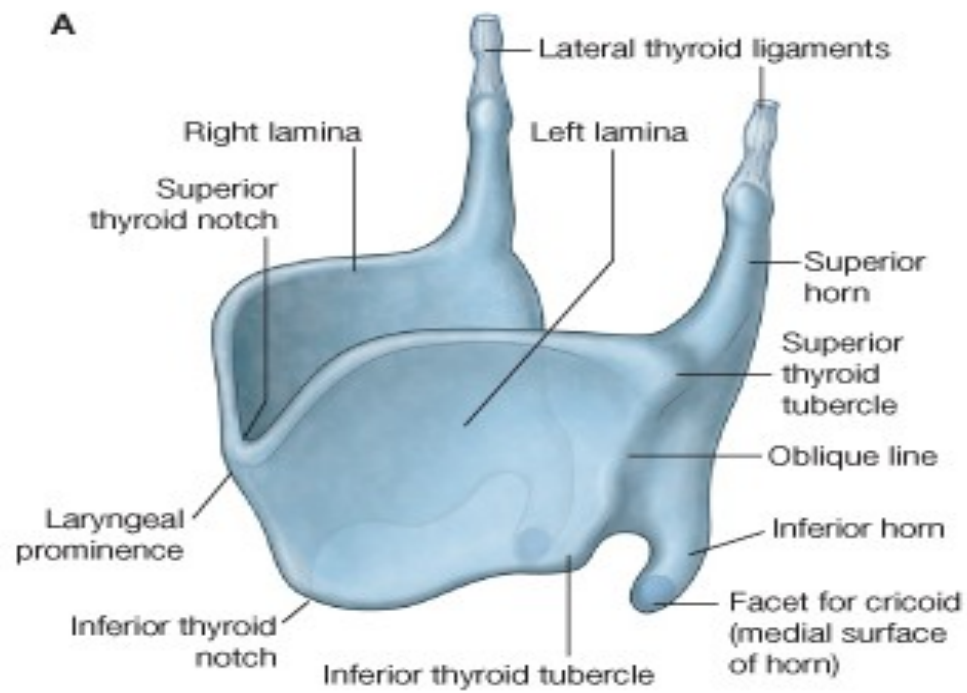
- Functions
 - Voice production
 - Air passage to trachea
 - Acts as a valve for preventing swallowed food from entering the lower respiratory tract

- Position
 - At midline of neck from root of tongue to trachea
 - In front of laryngopharynx opposite C3 to C6
 - Superiorly continuous with laryngopharynx through the inlet of larynx
 - Inferiorly continuous with trachea
 - 5 cm in adult man but slightly shorter in female & children

- **Structures**
 - 9 cartilages joined by ligaments and membranes
 - **Three unpaired** – thyroid, cricoid & epiglottis
 - **Three paired** – arytenoid, corniculate & cuneiforms
 - Muscles



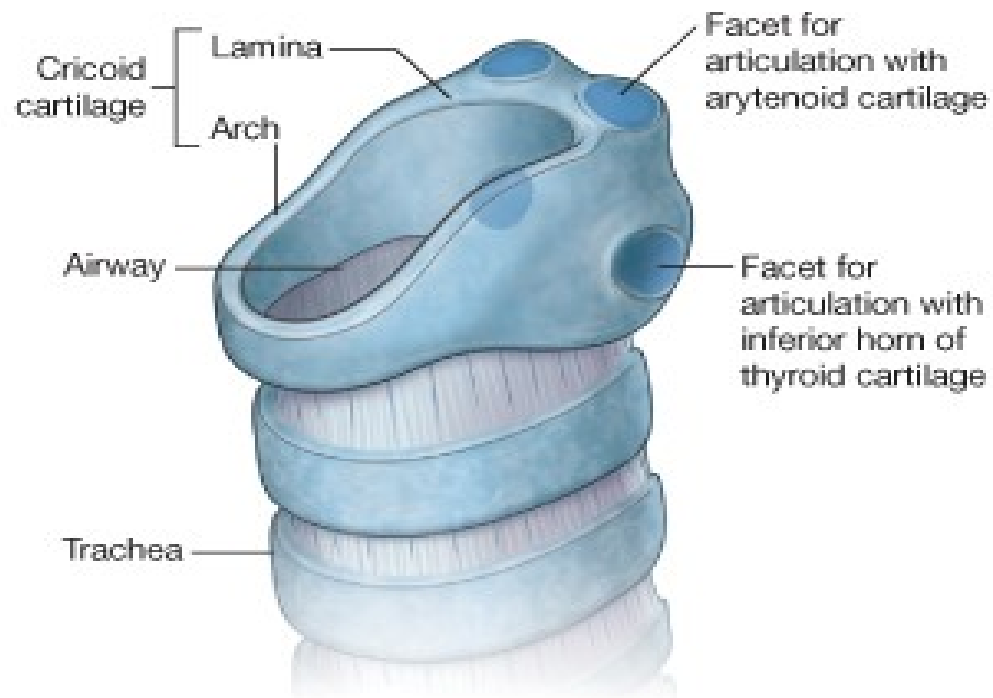




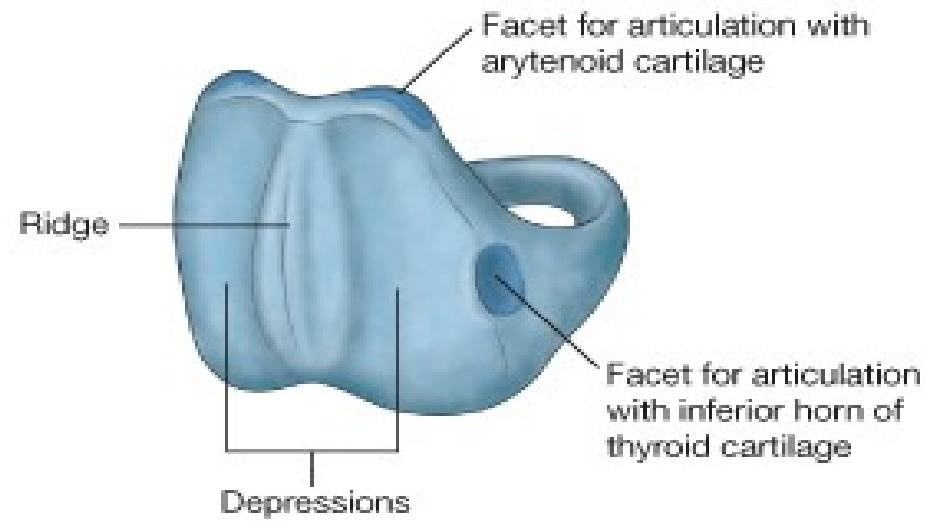
- **The cricoid cartilage**

- Ring shaped, having quadrilateral lamina posteriorly and narrow arch anteriorly
 - Lamina projects upwards and articulates with arytenoid
- is a key landmark in the neck, indicates the:
 - Level of the C6 vertebra.
 - Site where the carotid artery can be compressed against the transverse process of the C6 vertebra.
 - Junction of the larynx and trachea.
 - Joining of the pharynx and esophagus.
 - Point where the recurrent laryngeal nerve enters the larynx.
 - Site that is approximately 3 cm superior to the isthmus of the thyroid gland.

A

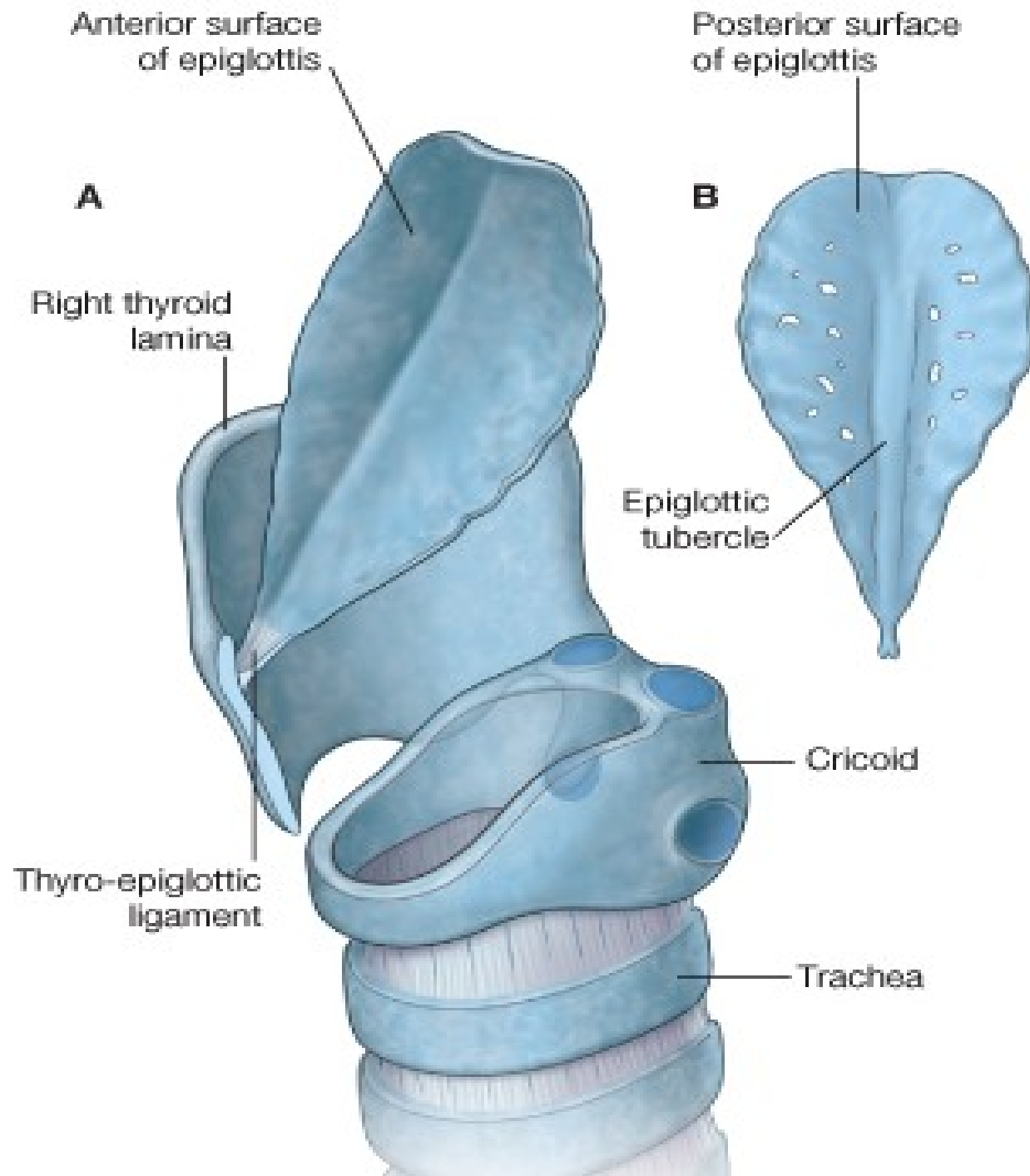


B

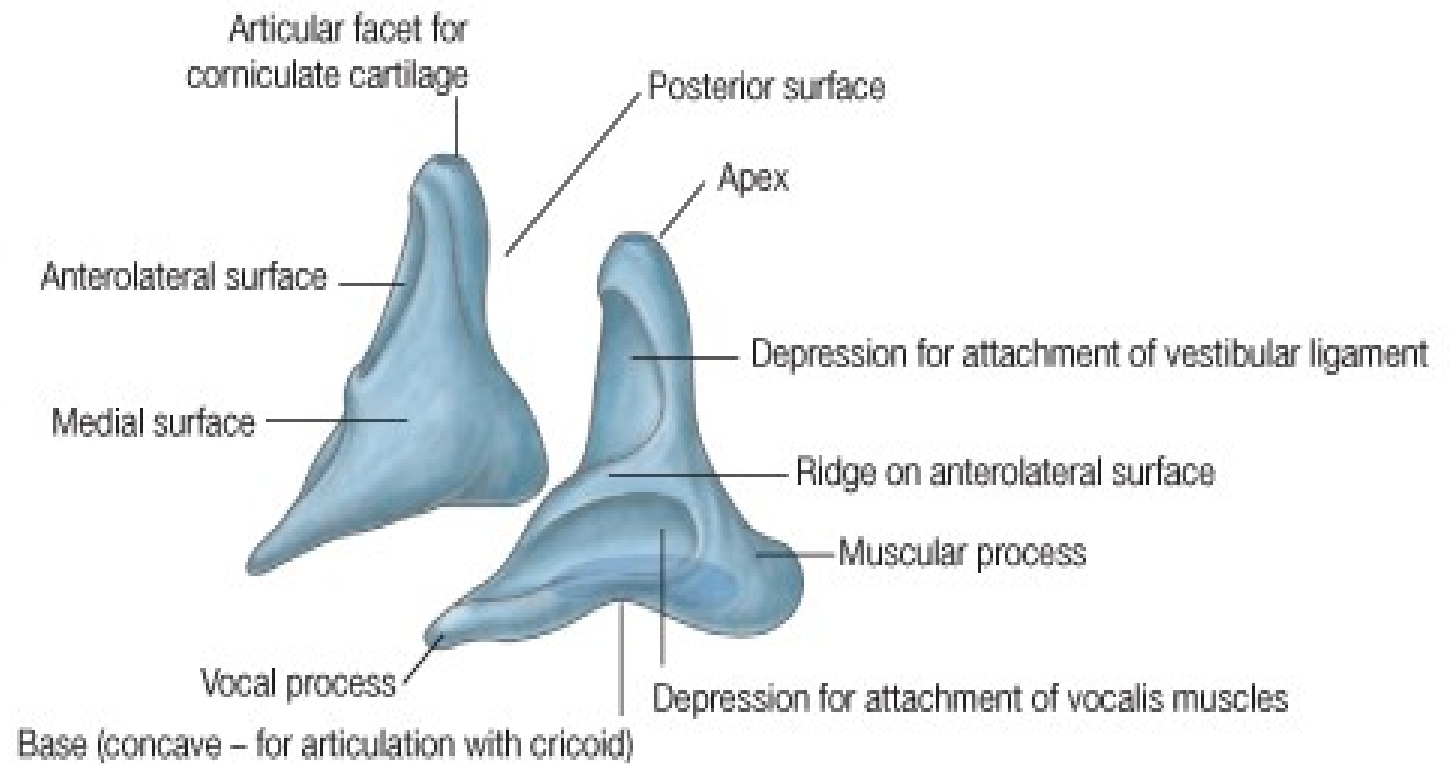


- **The epiglottis**

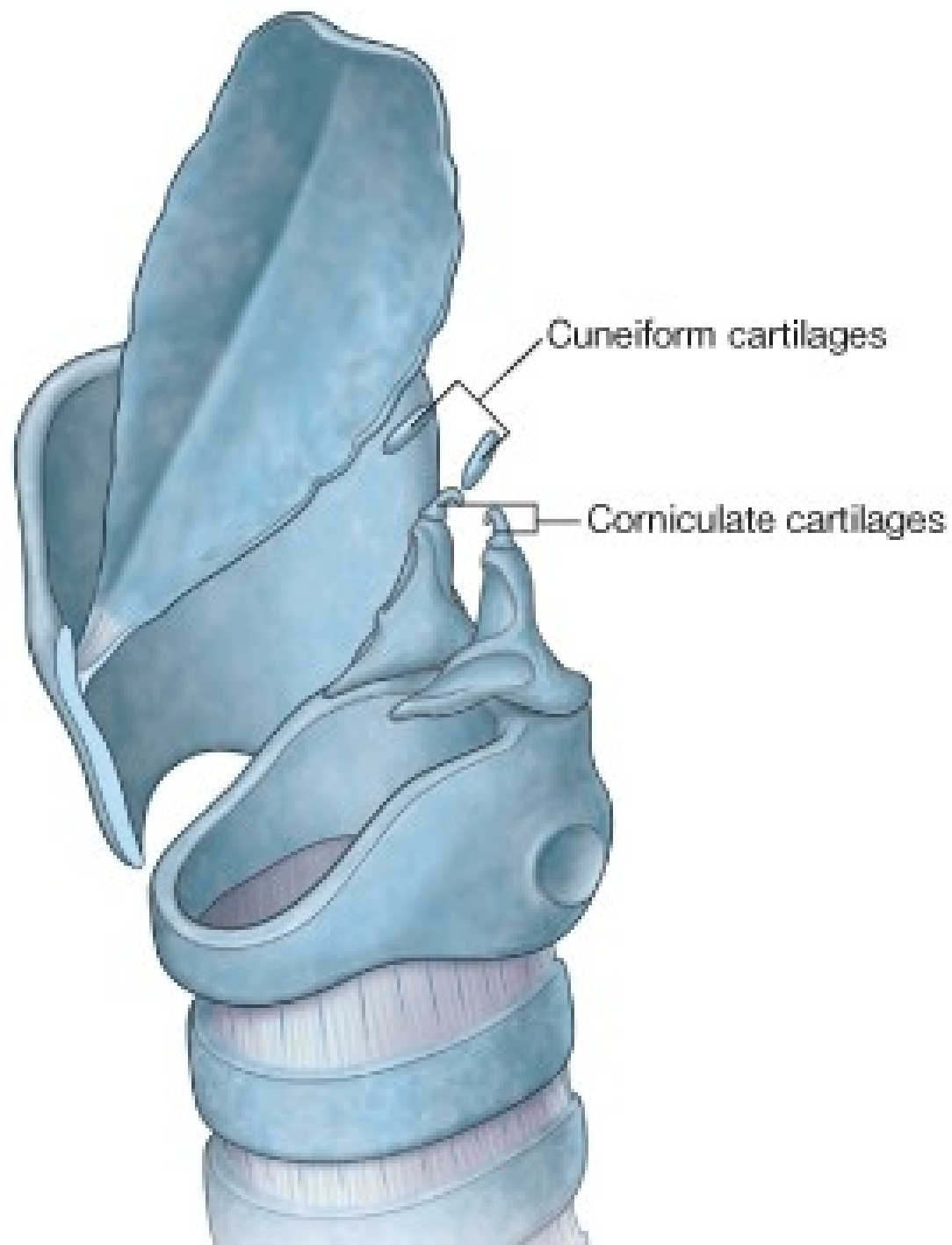
- leaf-shaped, broad flat plate of fibrocartilage at anterior wall of upper part of larynx
- **upper end** – broad, free, projects behind the tongue and hyoid bone, above and in front of the inlet of larynx
- **lower end** – pointed, attached above the angle of fusion of the thyroid cartilage by the thyroepiglottic ligament
- anterior surface is connected to tongue by glossoepiglottic fold and hyoid by hyoepiglottic ligament



- **The arytenoid cartilages**
 - Three sided pyramid with a base and apex
 - On upper border of lamina of cricoid
 - **Apex** – curved posteromedially and articulate with corniculate cartilage
 - **Base** - articulate with cricoid lamina through synovial joint called crico-arytenoid
 - **Has two processes**
 - **vocal process** - anterior & attach to the vocal ligament
 - **muscular process**- lateral & serves as attachment points for muscles of the larynx
 - control the opening between the vocal cords, the rima glottidis



- **Corniculate**
 - Small conical nodules of fibrocartilage, articulating with the apex of arytenoid cartilage in the posterior part of aryepiglottic fold
- **Cuneiform**
 - Rod shaped, found in the aryepiglottic fold, anterior to the arytenoid cartilage



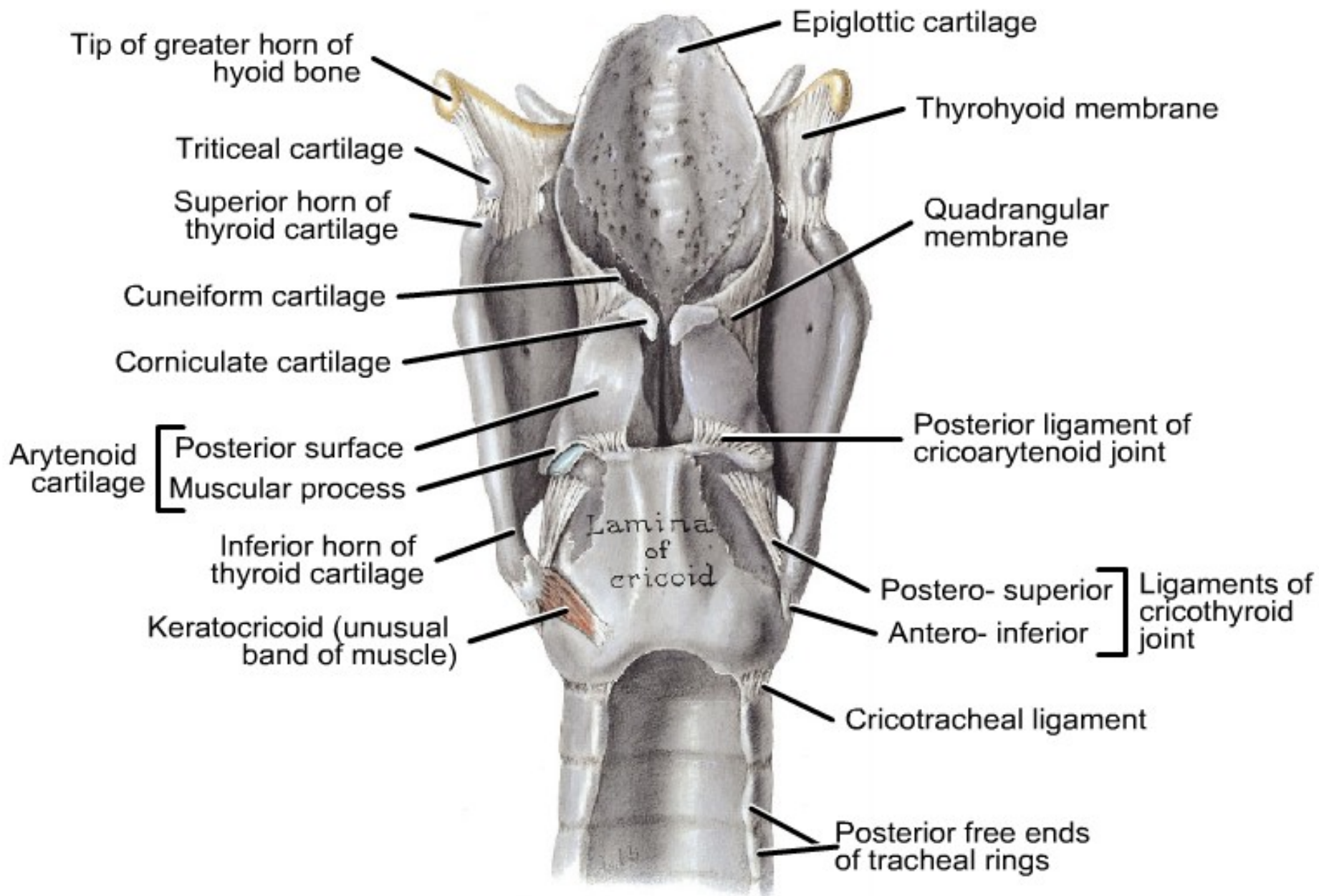
Joints of larynx

- **Crico-arytenoid**

- Synovial, between base of arytenoid and upper border of lamina of cricoid cartilage
- Permits sliding, tilting and rotary movements of arytenoids which are important in approximating, tensing and relaxing vocal folds

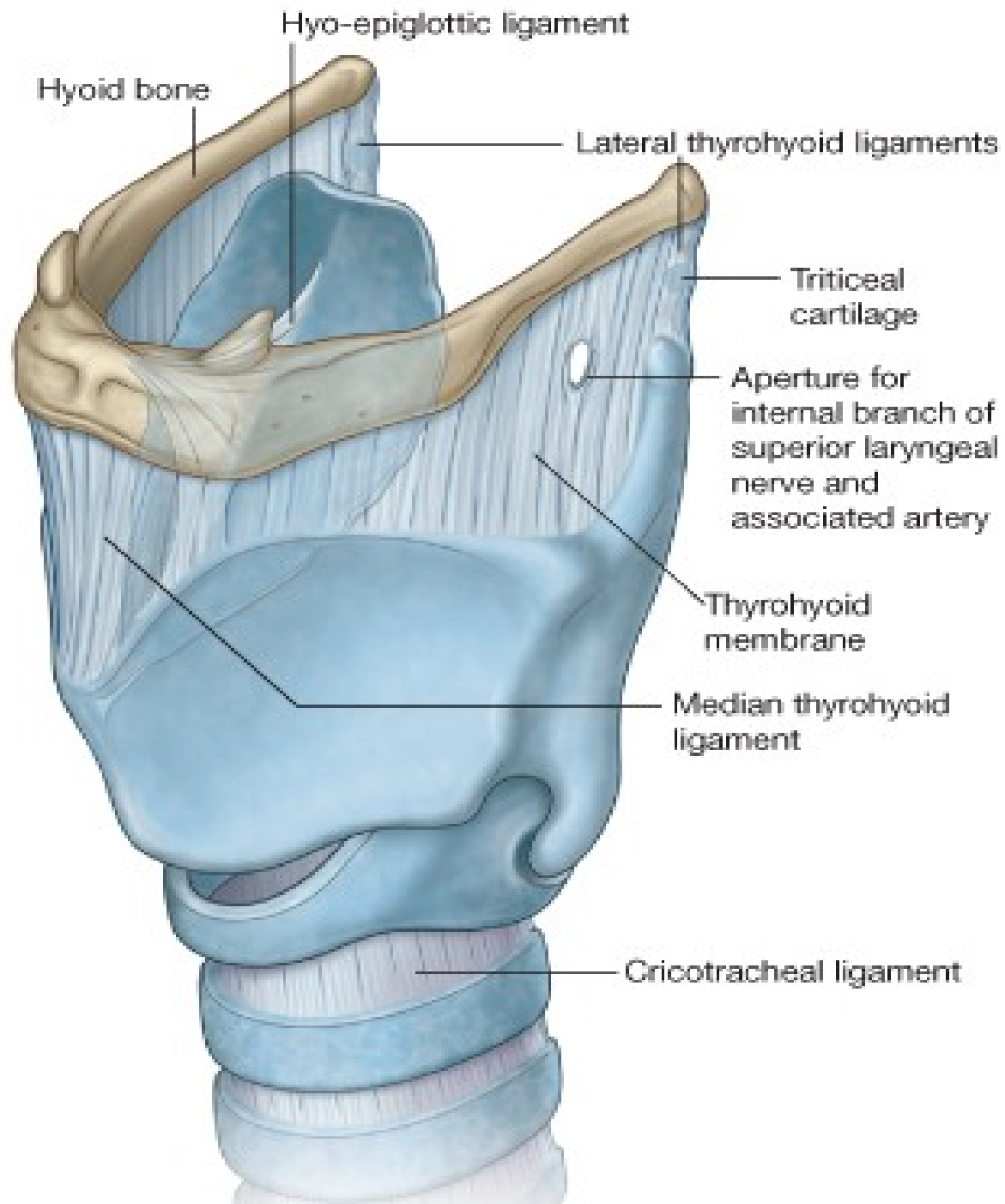
- **Crico-thyroid**

- Synovial, between inferior horn of thyroid cartilage and the sides of cricoid cartilage at the junction of its lamina and arch
- Permits rotary and gliding movements which permit changes in length of vocal folds



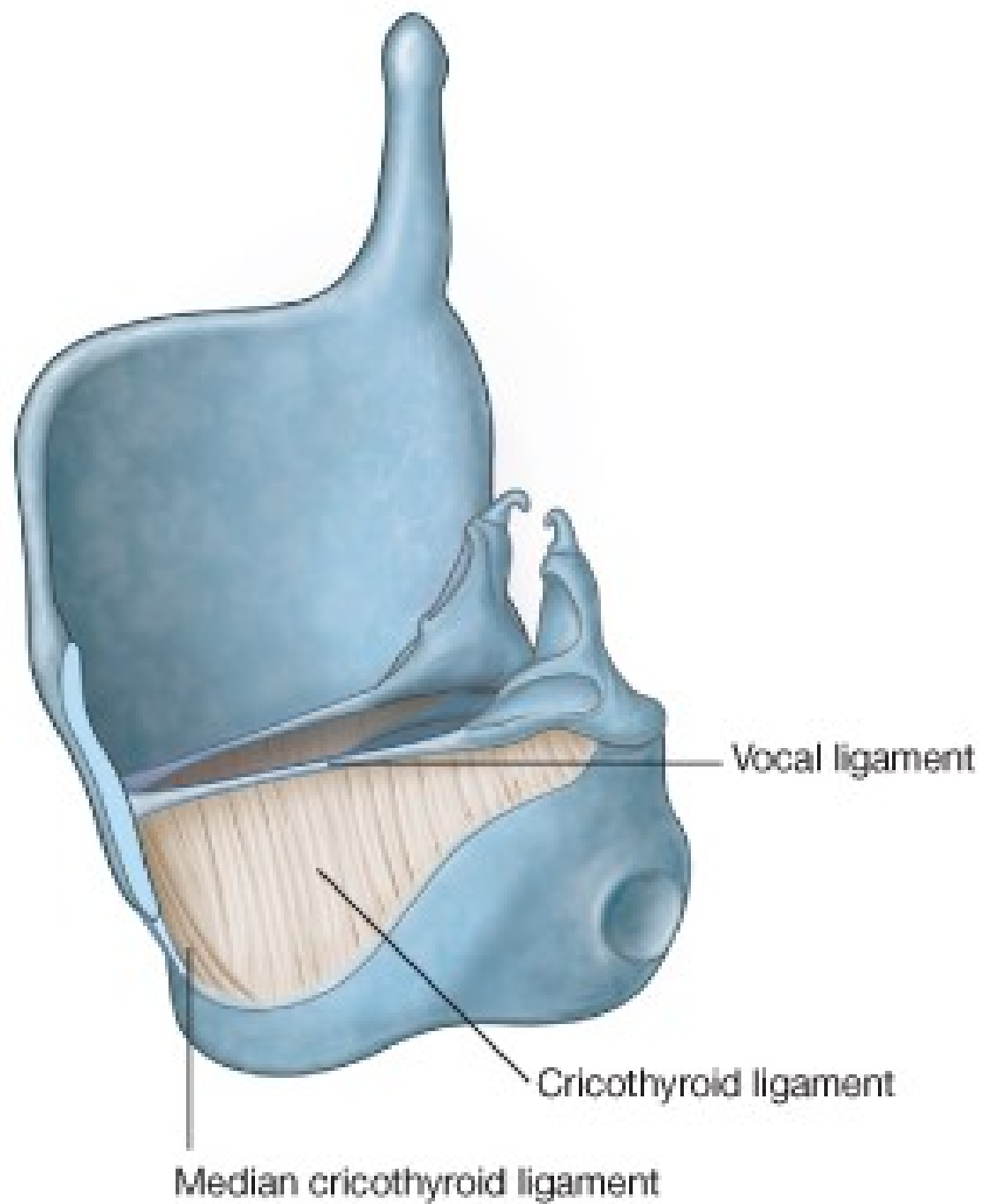
Ligaments and membranes

- **Thyrohyoid membrane & ligament**
 - Connect upper border of thyroid cartilage to body, greater & lesser horn of hyoid
 - Midline thickening – median thyrohyoid ligament
 - Posterior border thickening – lateral thyrohyoid ligament
 - Outer surface covered with thyrohyoid muscle
 - Pierced by superior laryngeal vessels & internal laryngeal nerve

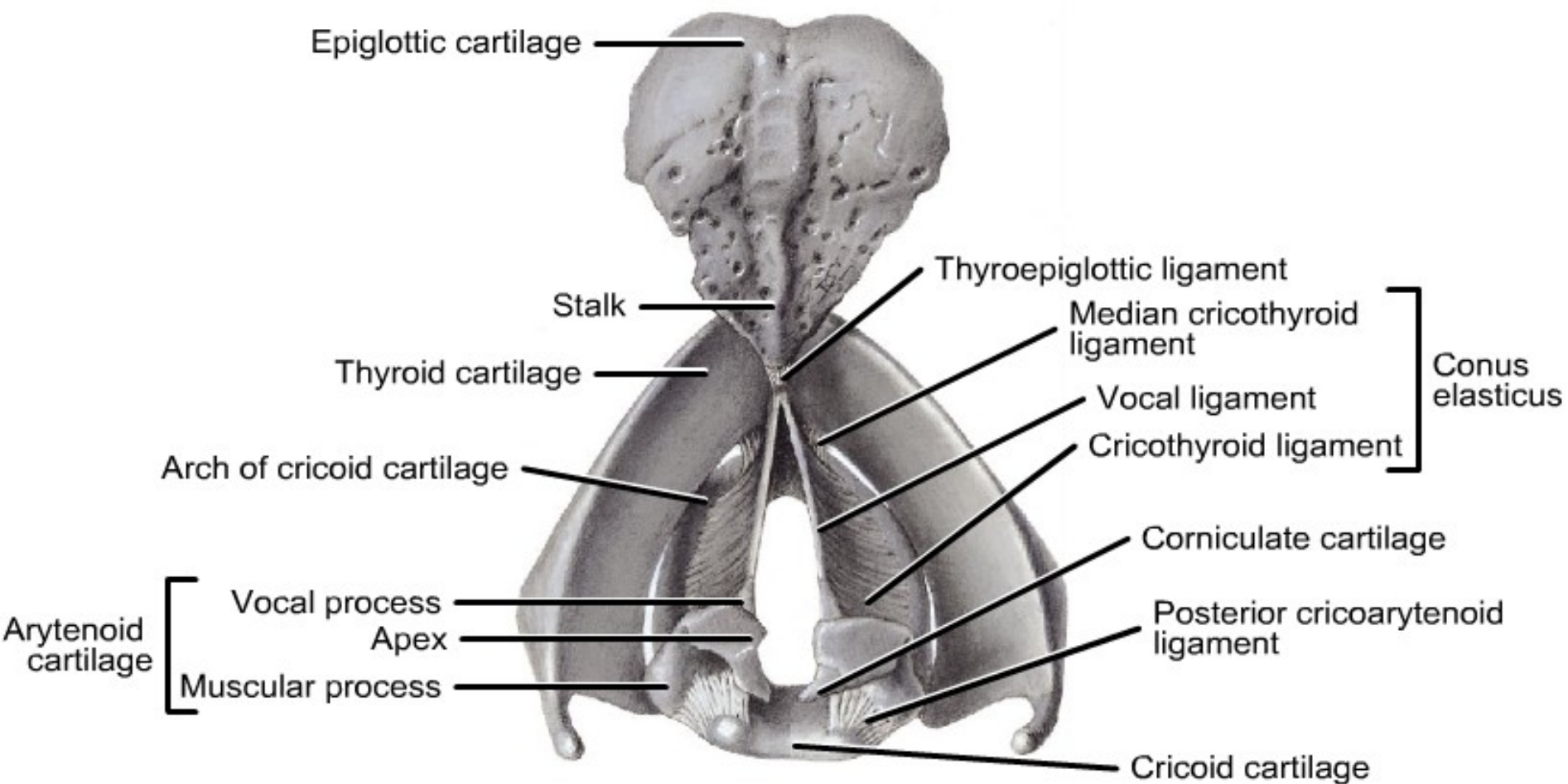


- **Cricothyroid ligament**

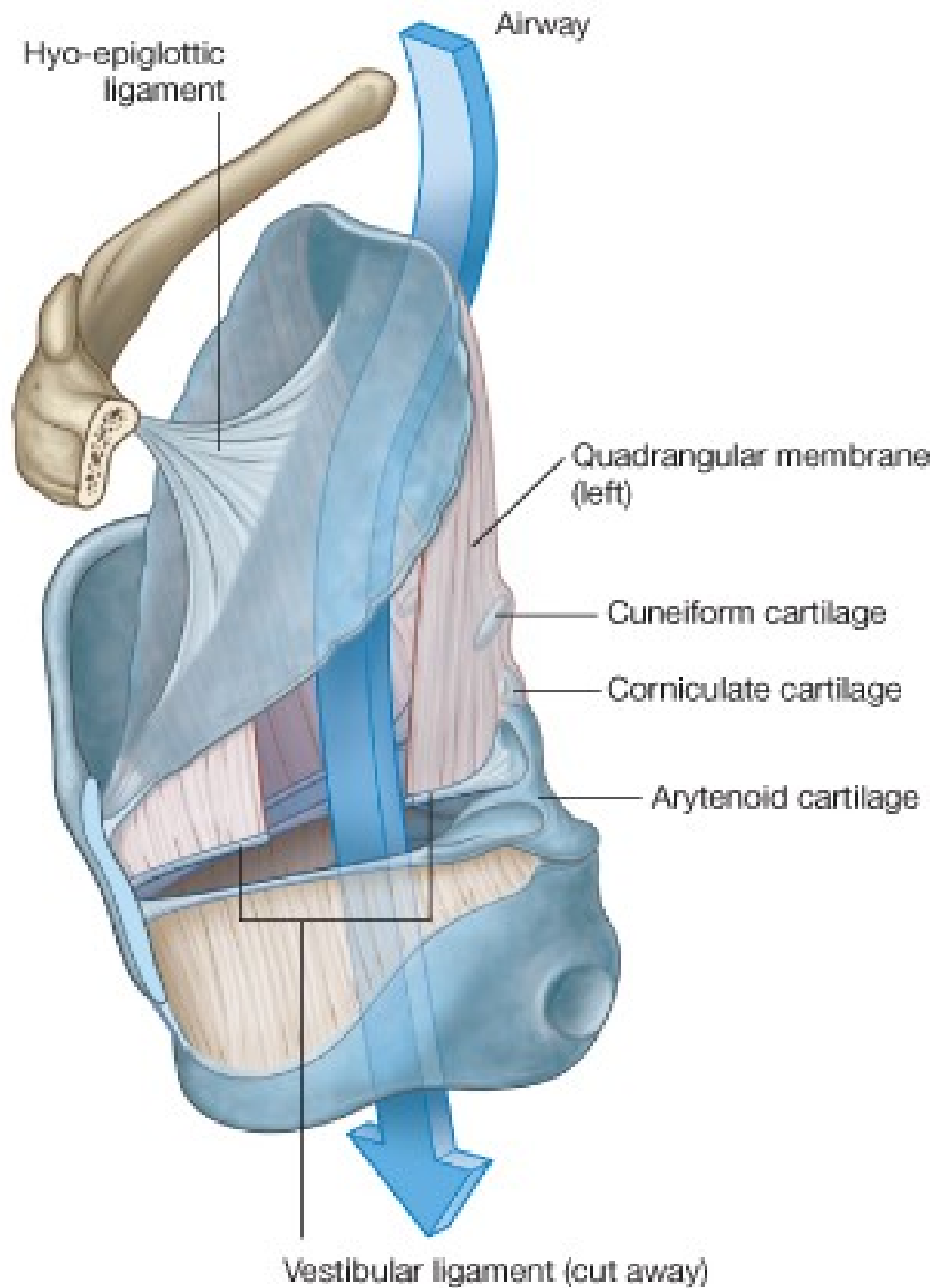
- Triangular in shape
- Connects upper border of arch of cricoid cartilage with deep surface of thyroid cartilage near the lower border
- Outer surface covered with cricothyroid muscle
- Inner surface covered with mucous membrane
- Soft spot inferior to thyroid cartilage - air way is closest to skin and most accessible

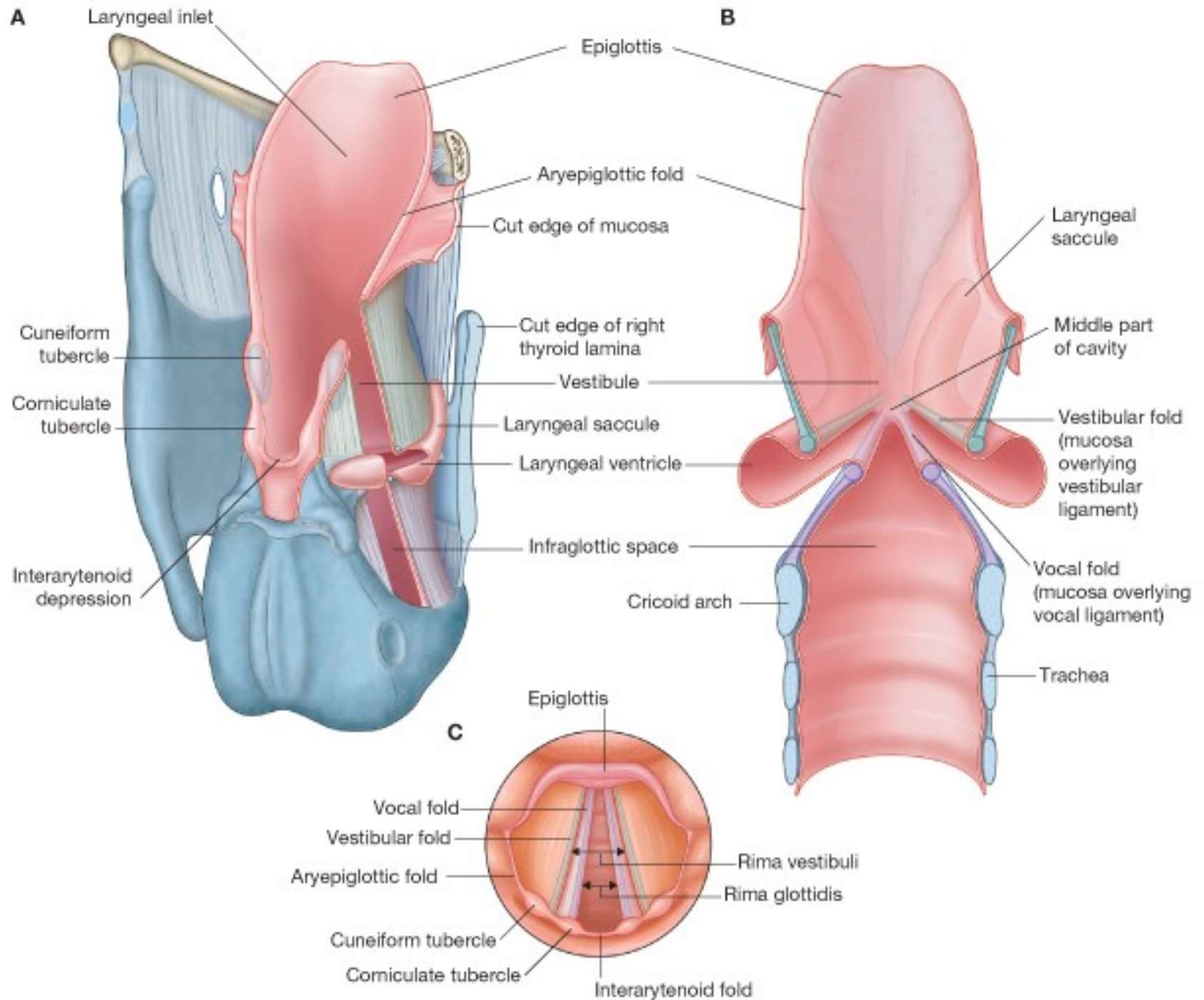


- **Vocal ligament, vocal fold & conus elasticus**
 - **Conus elasticus**
 - triangular ligament connecting cricoid, thyroid & arytenoid cartilage
 - **Vocal ligaments**
 - attach between the vocal processes of the arytenoid cartilages and the thyroid cartilage, behind its fusion angle
 - i.e. thickened upper border of conus elasticus
 - forms skeleton of vocal folds
 - **Vocal fold**
 - the upper border of conus elasticus covered with the mucous membrane



- **Quadrate membrane & vestibular ligaments**
 - **Quadrate membrane**
 - Thin submucosal sheath extending from arytenoid cartilage to the epiglottis
 - Superior border forms aryepiglottic fold
 - Inferior border forms vestibular fold
 - **Vestibular ligaments**
 - The free inferior margin the quadrate membrane
 - Covered with mucous membrane called vestibular fold (superior to the vocal fold)
 - Extends from the arytenoid to the thyroid cartilage





Quadrangular membrane

Epiglottis

Vestibular
ligament

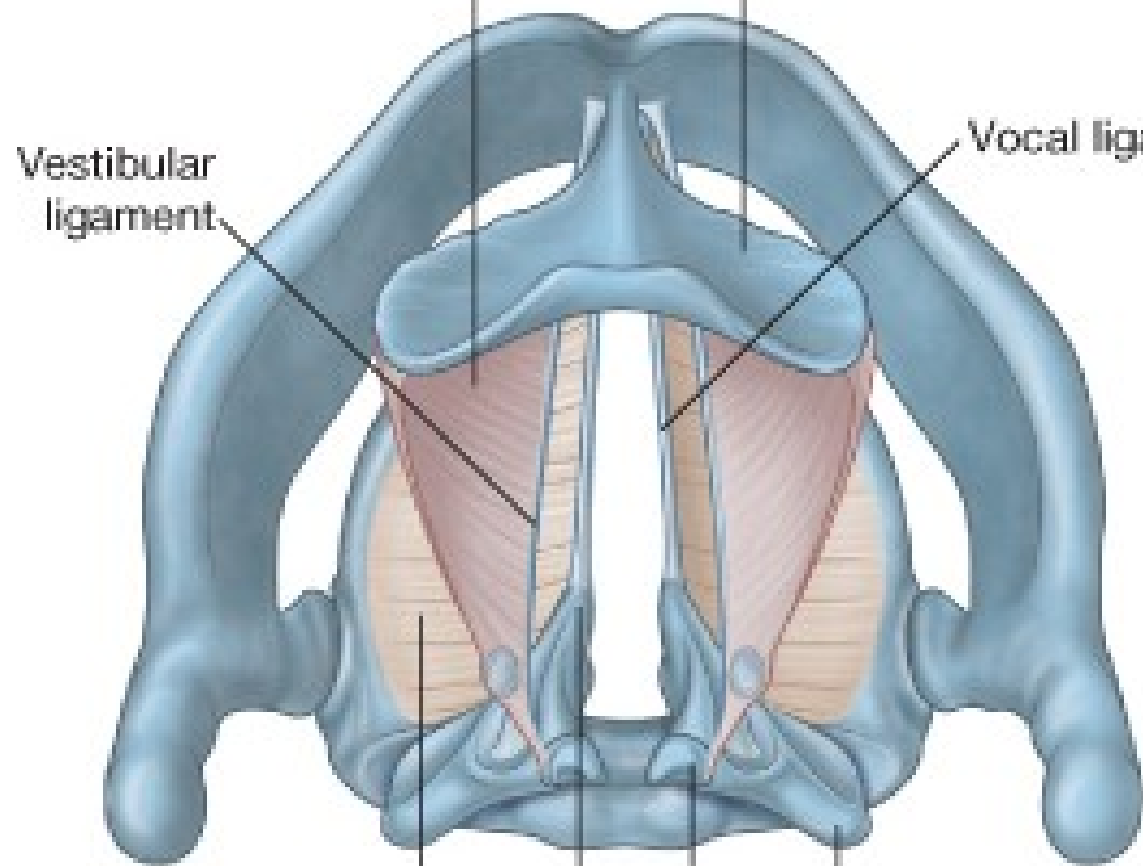
Vocal ligament

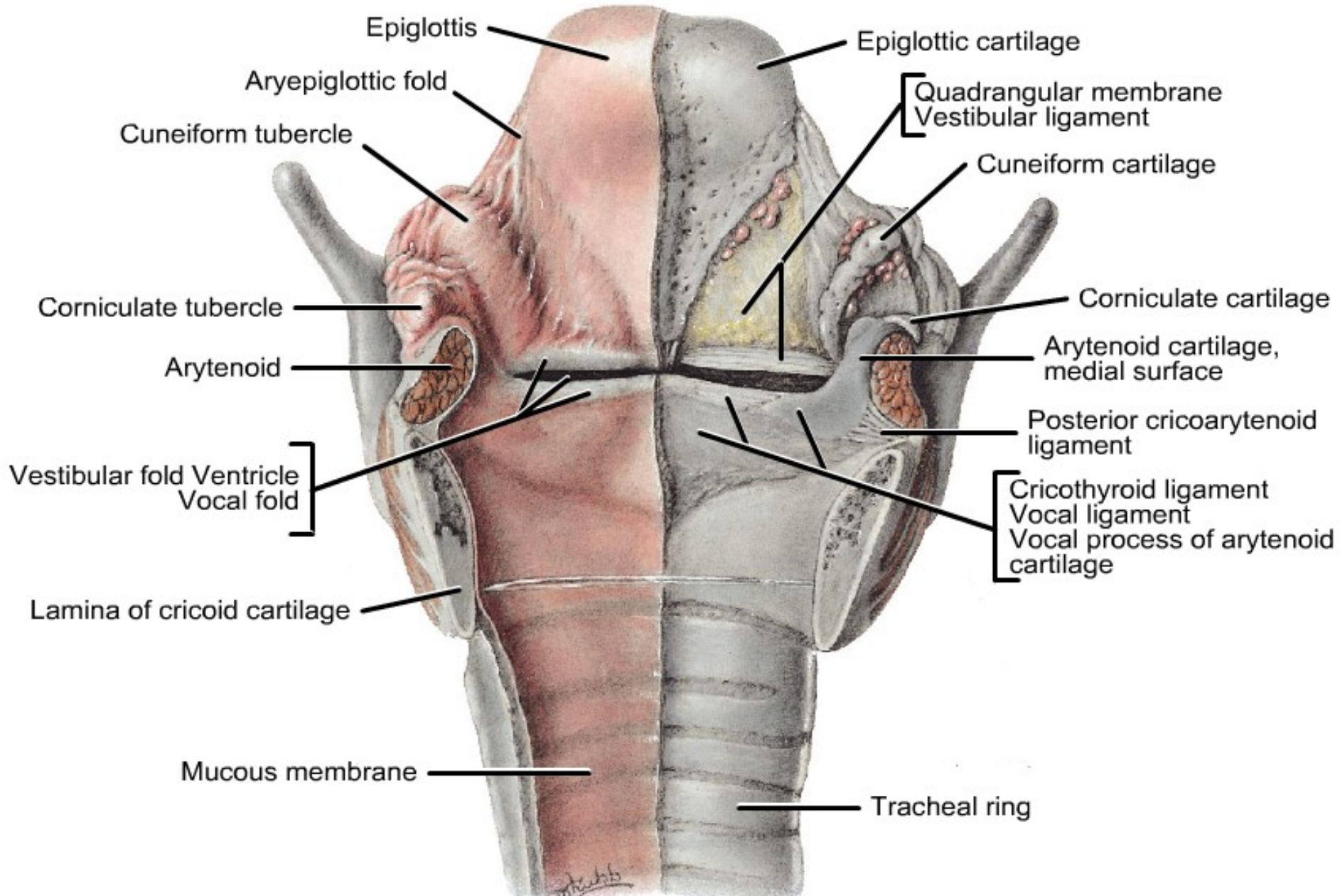
Cricothyroid ligament

Muscular process
of arytenoid

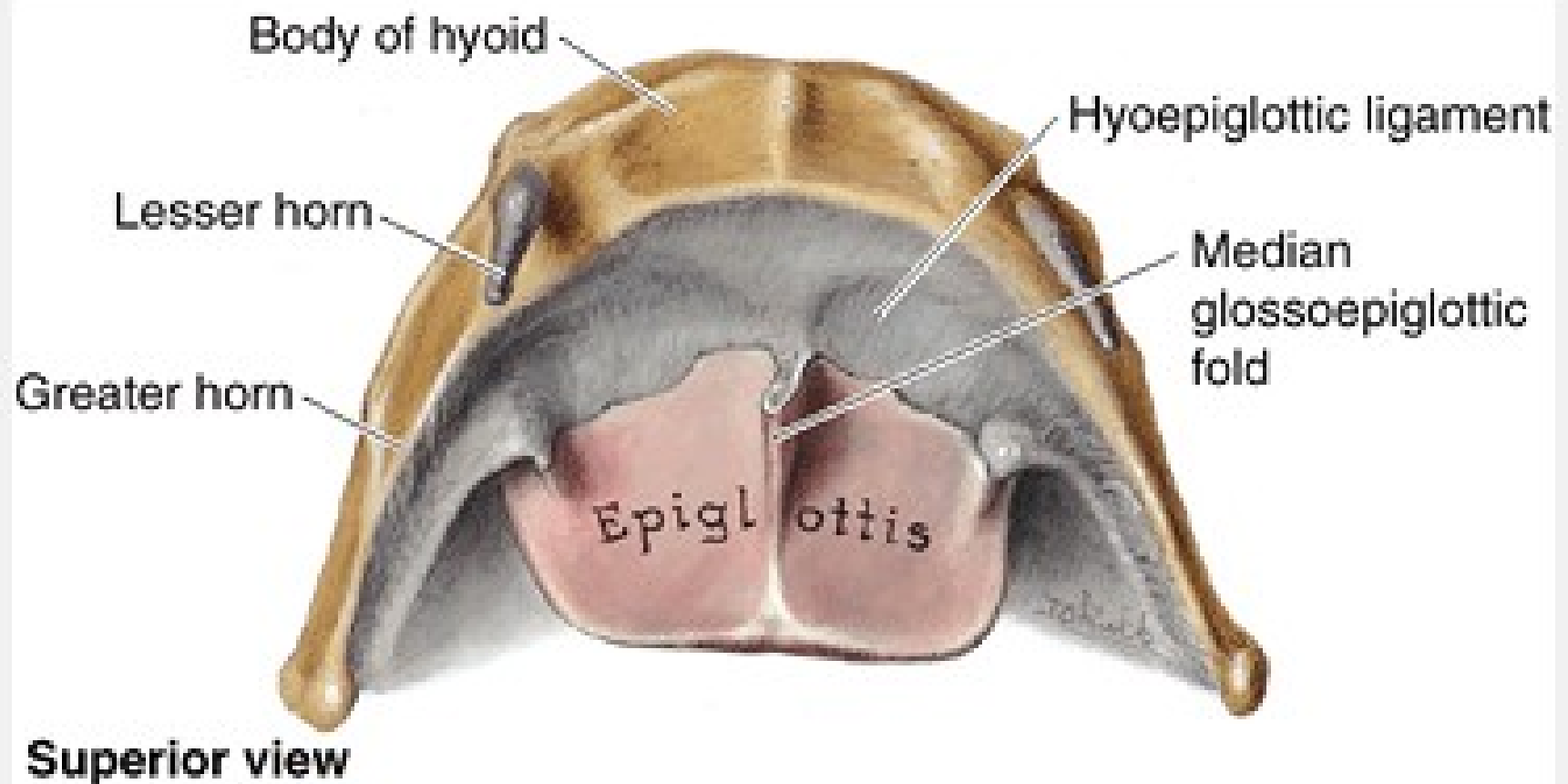
Vocal process of arytenoid

Comiculate cartilage





- **Hyoepiglottic ligament**
 - Connects the anterior surface of the epiglottis to the back of hyoid
- **Thyro-epiglottic ligament**
 - Lower end of the epiglottis to the back of thyroid
- **Ary-epiglottic folds of mucus membrane**
 - Connects the sides of the epiglottis with arytenoid cartilage
- **Median & lateral glosso-epiglottic folds**
 - Connects the anterior surface of the epiglottis to the dorsum the tongue
- **Crico-tracheal ligament**
 - Connects first ring of trachea to cricoid



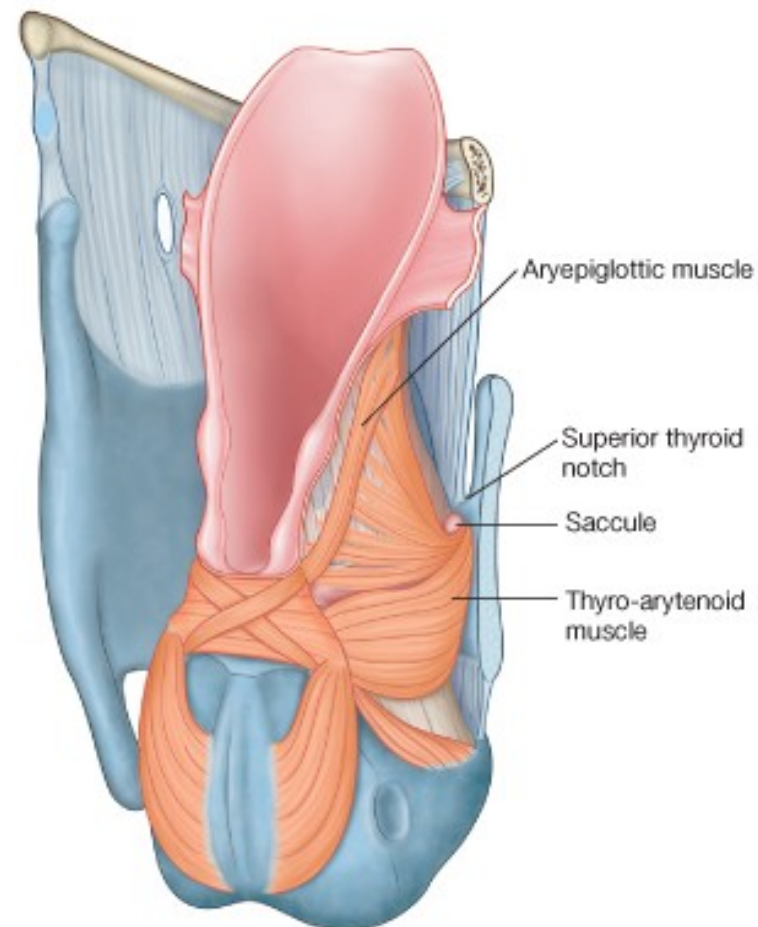
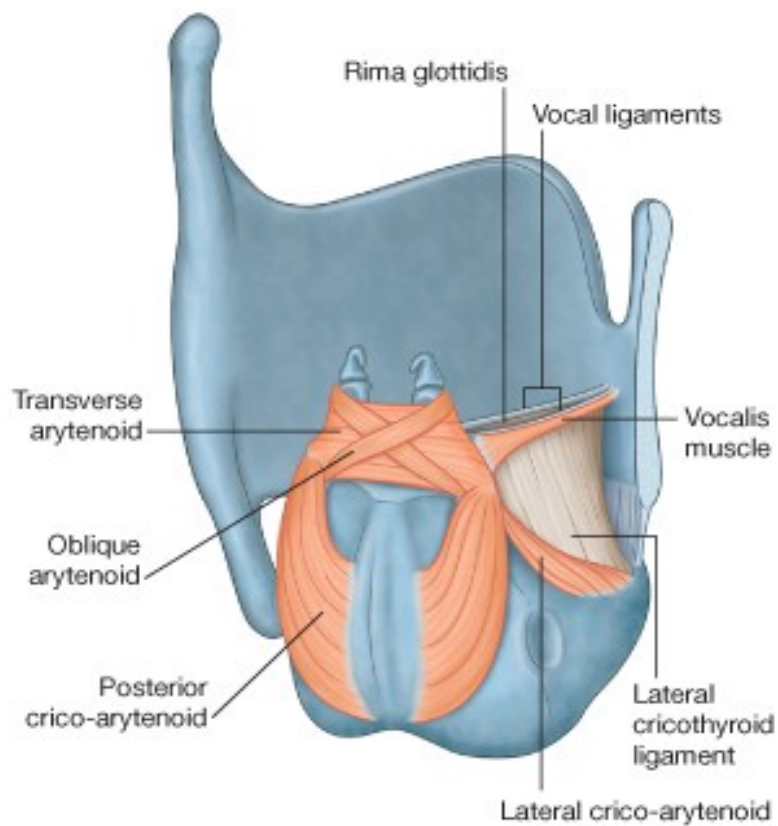
Muscles of larynx

- **Extrinsic muscles** – move larynx during swallowing
 - Infrahyoid muscles – depress hyoid and larynx
 - Suprahyoid muscles – elevates hyoid and larynx
 - Thyrohyoid – depress hyoid and elevate thyroid cartilage
- **Intrinsic muscles** – move laryngeal parts
 - Alter length and tension of vocal folds and size and shape of rima glottidis
 - Divided based on their action on inlet of larynx

Muscles of larynx

- Muscles that abduct or adduct the vocal cords
 1. **Posterior cricoarytenoids:** abduct the vocal cords.
 2. **Lateral cricoarytenoids:** adduct the vocal cords.
 3. **Transverse arytenoid:** adduct the vocal cords.
- Muscles that increase or decrease the tension of vocal cords
 1. **Cricothyroid:** tenses the vocal cords.
 2. **Vocalis:** tenses the vocal cords.
 3. **Thyroarytenoid:** relaxes the vocal cords.

- **Muscles closing the inlet of the larynx**
 - **Transverse arytenoid muscle**
 - unpaired
 - attaches to the posterior aspects of the 2 arytenoid cartilage
 - Its contraction will adduct the arytenoid cartilages, vocal processes and vocal cords
 - **Oblique arytenoid**
 - Superficial to transverse
 - 2 bundles cross each other, each arises from the muscular process of one arytenoid to the apex of the other arytenoid



- **Abductor of vocal folds**

- **Posterior crico-arytenoid muscle**

- From the posterior surface of the lamina of the cricoid cartilage to the posterior aspect of the muscular process of the arytenoid cartilage
 - Rotate arytenoids thus abduct the vocal processes and open the rima glottidis

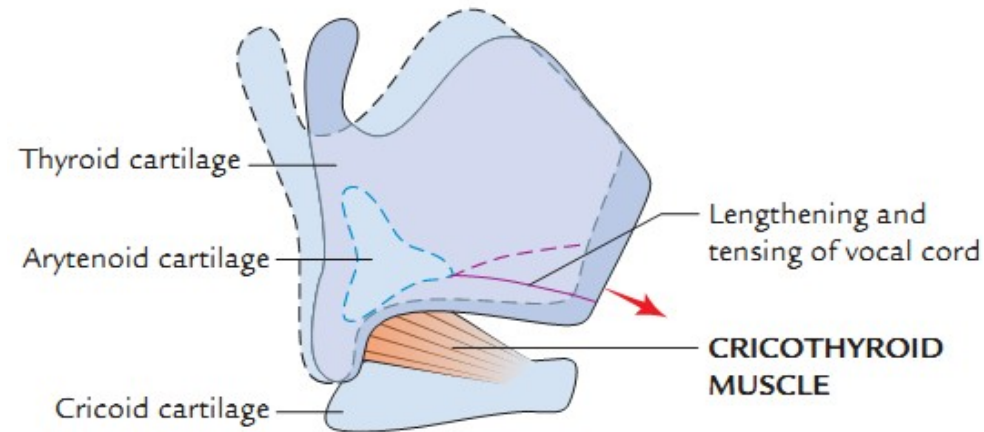
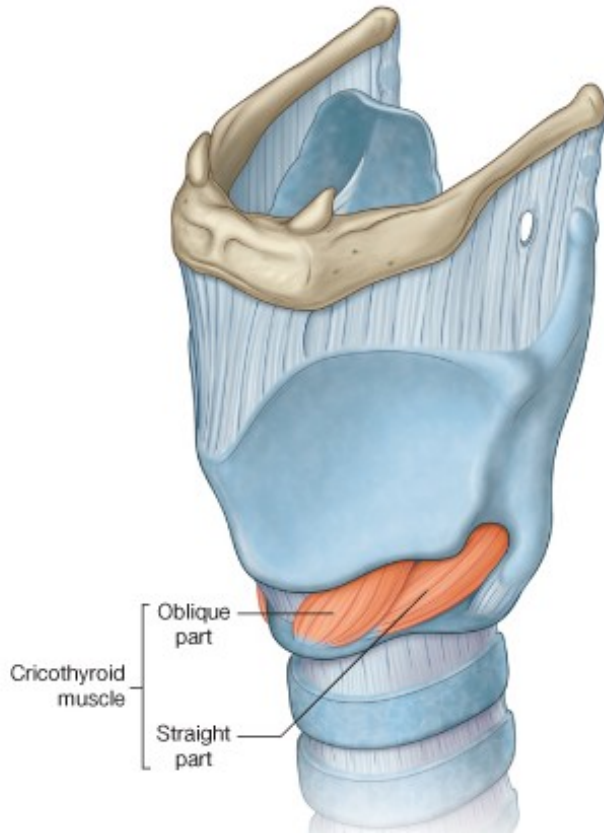
- **Thyro-epiglottic muscle**

- Connect the deep surface of lamina of thyroid cartilage with sides of the epiglottis
 - Superior fibers of the thyroarytenoid muscles pass into the aryepiglottic fold, and some of them reach the epiglottic cartilage; These fibers constitute the thyroepiglottic muscle
 - Pulls the epiglottis fore ward and widens the inlet

- **Tensor of vocal cords**

- **Cricothyroid muscle**

- Arise from lower border of the cricoid and insert into the lower border of the inferior horn of thyroid
 - Its contraction tilts the thyroid cartilage g in tension of the vocal cords

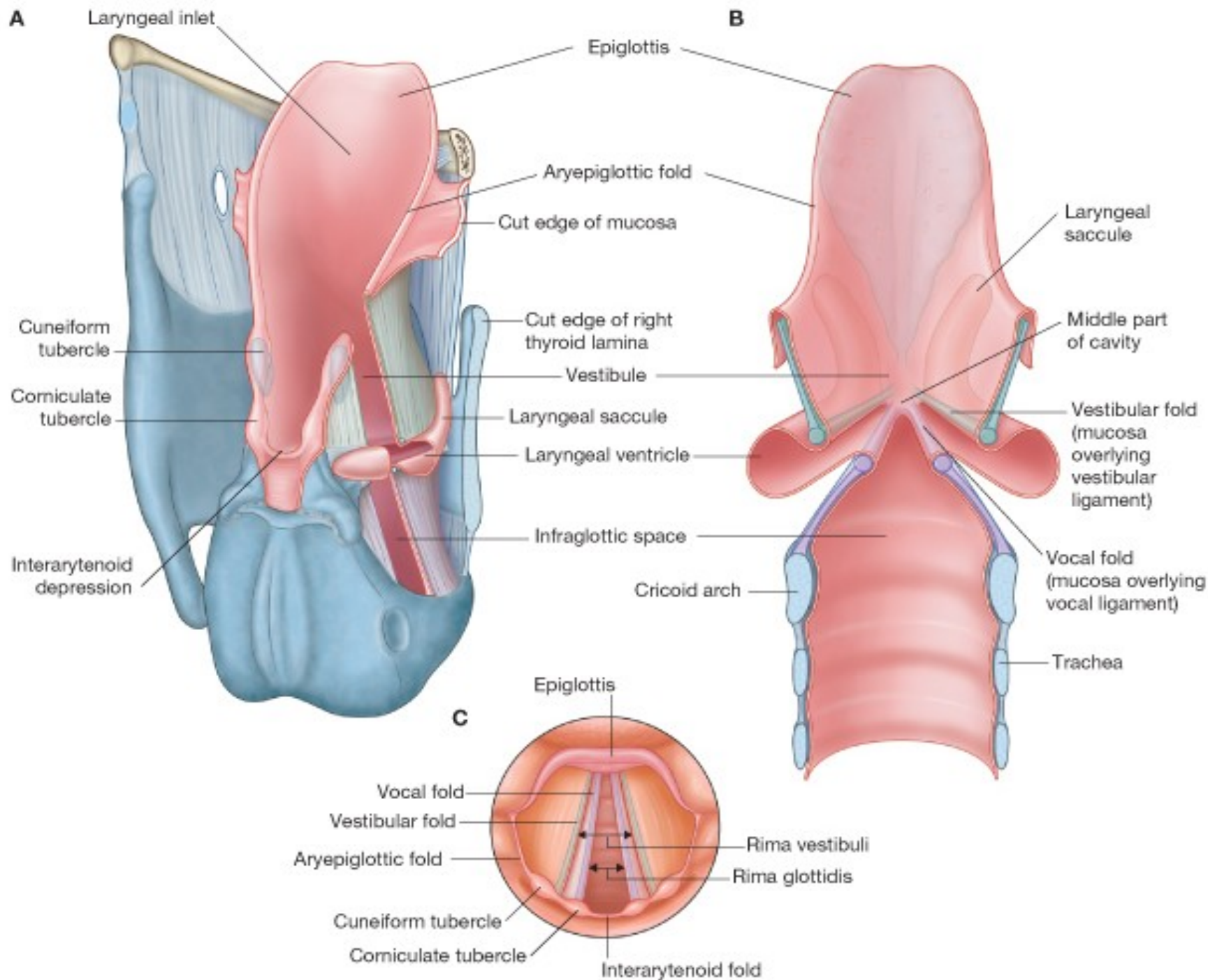


- **Adductors of vocal folds**
 - **Lateral crico-arytenoid muscle**
 - Attaches from the superior aspect of the arch of the cricoid to the anterior surface of the muscular process of the arytenoid cartilage
 - Rotate the arytenoid cartilages medially and approximate the vocal folds

- **Relaxer of vocal folds**
 - **Thyro-arytenoid muscle**
 - Arise from the internal surface of the angle of the thyroid cartilage and insert into the lateral aspect of the arytenoid cartilage
 - Pull arytenoid anteriorly thus relax vocal ligaments
 - The portion adjacent to the vocal ligament is the vocalis muscle

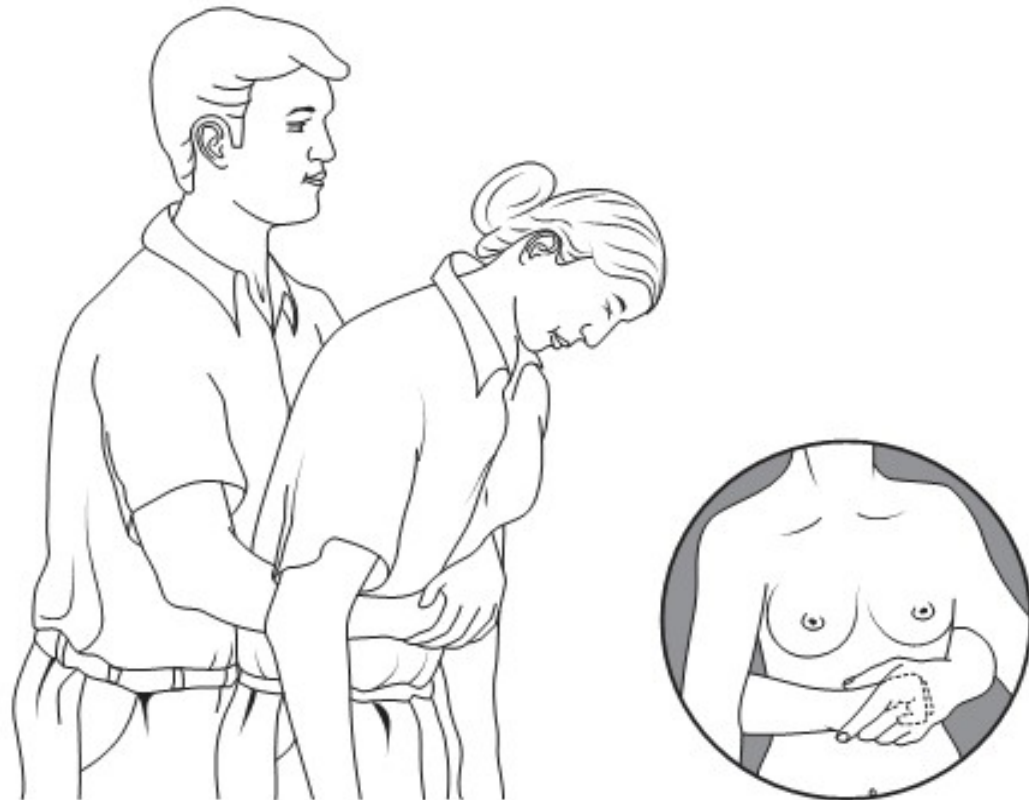
- **Cavities of the larynx**

- Extend from inlet of larynx to inferior border of cricoid
- Superior laryngeal aperture (inlet) is obliquely oriented and bounded by the aryepiglottic folds (laterally), interarytenoid fold (posterior) and epiglottis (anteriorly)
- Two folds of mucous membrane – vestibular (upper) and vocal (lower) folds
 - **Rima vestibuli** – spaces between the vestibular folds (false vocal cords)
 - **Rima glottidis** – spaces between the vocal folds
- The two folds divide the cavity into 3 parts
 - **Vestibule** – superior to vestibular fold
 - **Ventricle** – between vestibular and vocal folds
 - **Infraglottic cavity** – below vocal folds



- **Vocal folds**
 - Concerned with production of sound
 - Wedge-shaped; apex pointed medially and base lies against thyroid lamina
 - Consists of vocal ligament, conus elasticus, muscle and mucous membrane
 - Vocal folds, rima glottidis and narrow part of larynx together are called glottis
 - Shape of rima glottidis vary according to position of vocal folds
 - Narrow during ordinary breathing
 - Wide during forced breathing
 - Linear slit during speaking

- **Clinical correlates**
 - **Laryngeal obstruction**
 - Lodging of aspirated food in rima glottidis
 - Compression of abdomen (Heimlich maneuver) expel air from lungs and dislodge the food



Blood supply

- Arteries
 - **Superior laryngeal artery** – branch of superior thyroid; runs with internal laryngeal nerve and pierce thyrohyoid membrane and supply interior of larynx up to vocal folds
 - **Inferior laryngeal artery** – branch of inferior thyroid; supply mucosa and muscles below vocal folds
- Veins
 - **Superior laryngeal vein** – drain into superior thyroid
 - **Inferior laryngeal vein** – drain into inferior thyroid

- **Innervation**

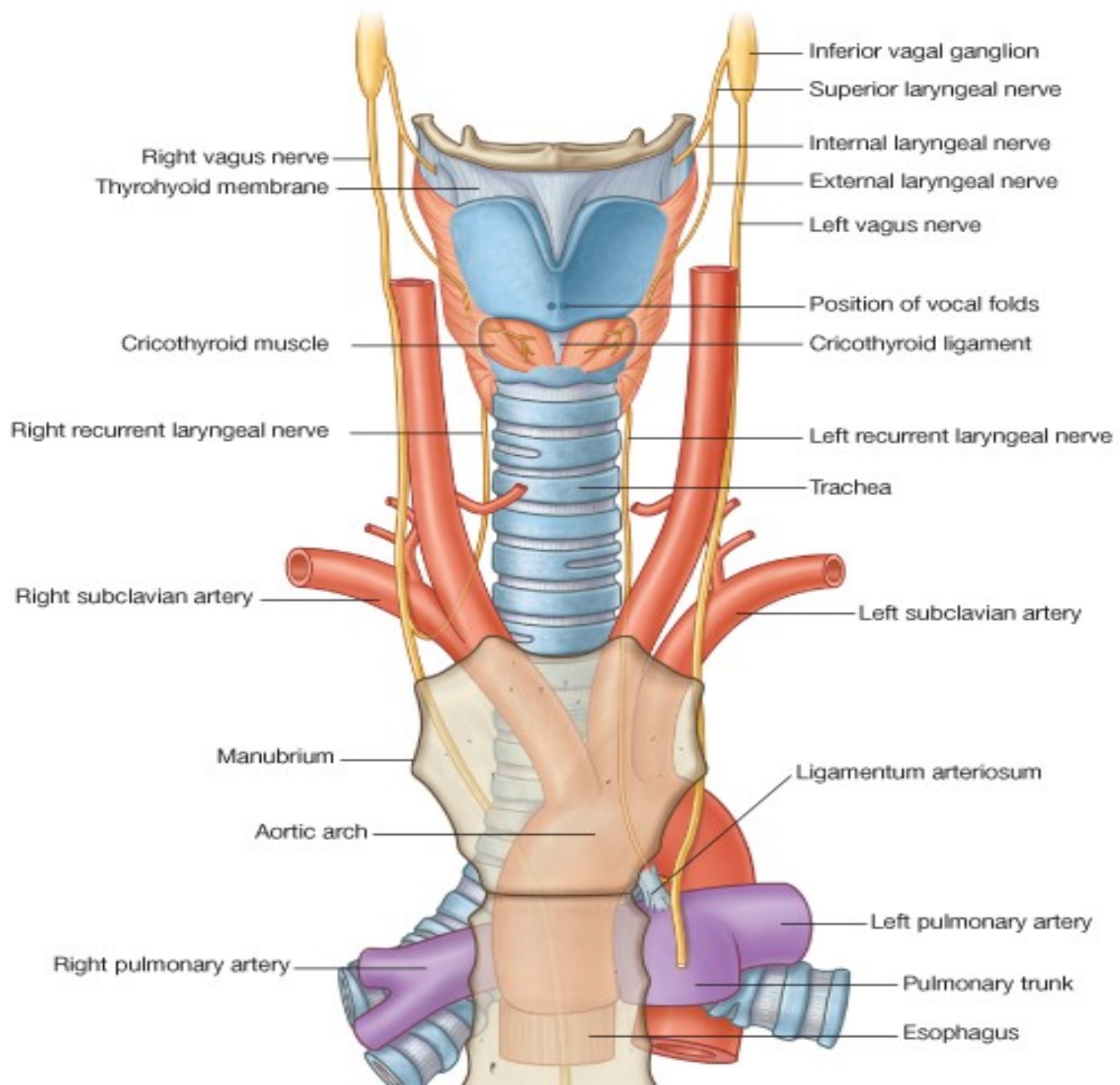
- **Motor**

- All intrinsic muscles are supplied by recurrent laryngeal nerve except cricothyroid (by external laryngeal)

- **Sensory**

- Internal laryngeal – above vocal folds
 - Recurrent laryngeal – below vocal folds

- **Superior laryngeal nerve**
 - A branch of vagus, divided into 2 terminal branches
 - **Internal laryngeal nerve** – pierces thyrohyoid membrane and supply laryngeal mucosa above vocal folds
 - **External laryngeal nerve** – descends posterior to sternothyroid and supply inferior constrictor and cricothyroid
- **Recurrent laryngeal nerve**
 - Ascends in a groove between trachea and esophagus and gives branches to pharynx, esophagus and trachea
 - Supplies all intrinsic muscles of larynx except cricothyroid and mucous membrane below vocal folds
 - Terminal part enters larynx below inferior constrictor and divide into anterior and posterior branches



- **Applied anatomy**

- **Recurrent laryngeal nerve** is vulnerable to injury during thyroidectomy resulting in aphonia or reduction of voice
 - Damage to internal laryngeal nerve produces anesthesia of mucosa superior to vocal folds which break the reflex arc causing explosive coughing when foreign body enters larynx
 - Damage to external laryngeal nerve causes weakness of phonation → monotonous voice
 - Interruption of **both** recurrent laryngeal nerves results in vocal folds lie in cadaveric position between abduction and adduction and thus **phonation is completely lost**
 - They cannot be adducted for phonation, nor can they be abducted for increased respiration, resulting in stridor (high-pitched, noisy respiration).
- tracheotomy may become essential.

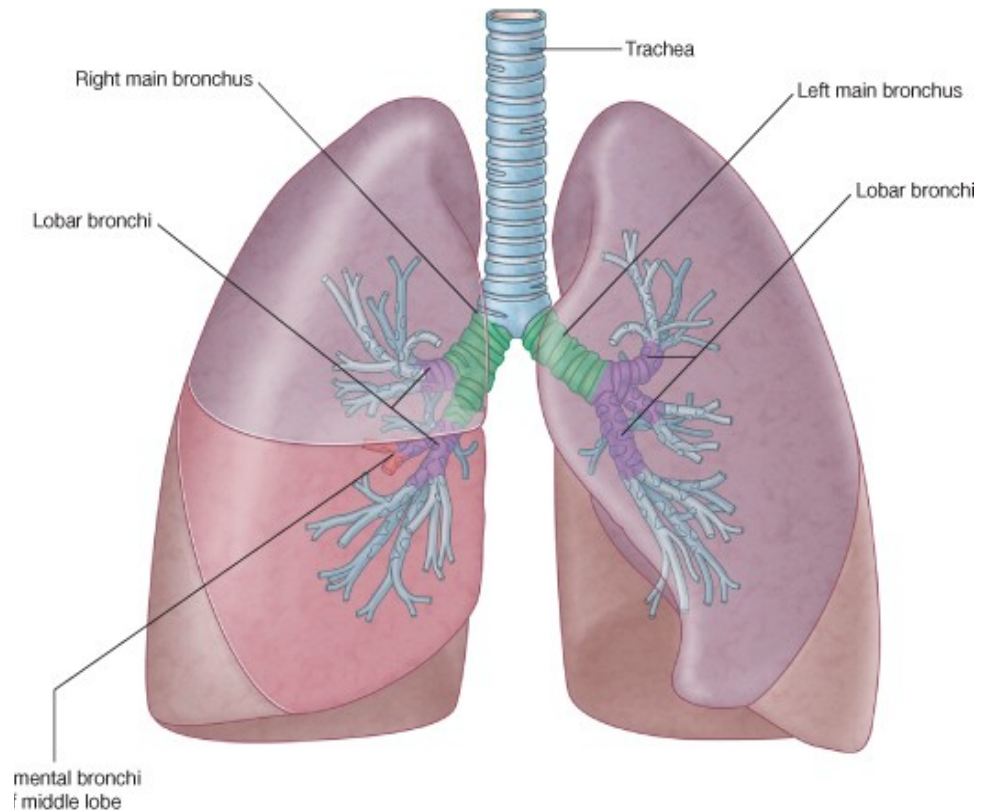
The laryngeal nerves

- laryngeal nerves have intimate relationships to the thyroid arteries: → practical importance in thyroidectomy.
- **External laryngeal nerve** lies immediately deep to superior thyroid artery & may be injured in ligating this vessel.
- **recurrent laryngeal nerve**, lying in the tracheo-oesophageal groove, is closely related with inferior thyroid artery.

.....*End*

Bronchopulmonary segment

- Each segmental bronchus, together with the part of the lobe it supplies, is called a **bronchopulmonary segment**.
- A ***bronchopulmonary segment*** is the zone of distribution of a ***segmental bronchus*** & is a functional, anatomical & surgical unit of lung tissue.
- ✂ ➔ Each **lobar (secondary) bronchus**, which passes to a lobe of the lung, gives rise to **segmental (tertiary) bronchi**
- ✂ ➔ each **segmental bronchus** then enters a **bronchopulmonary segment**.
- There are 10 ***bronchopulmonary segments*** in each lung.



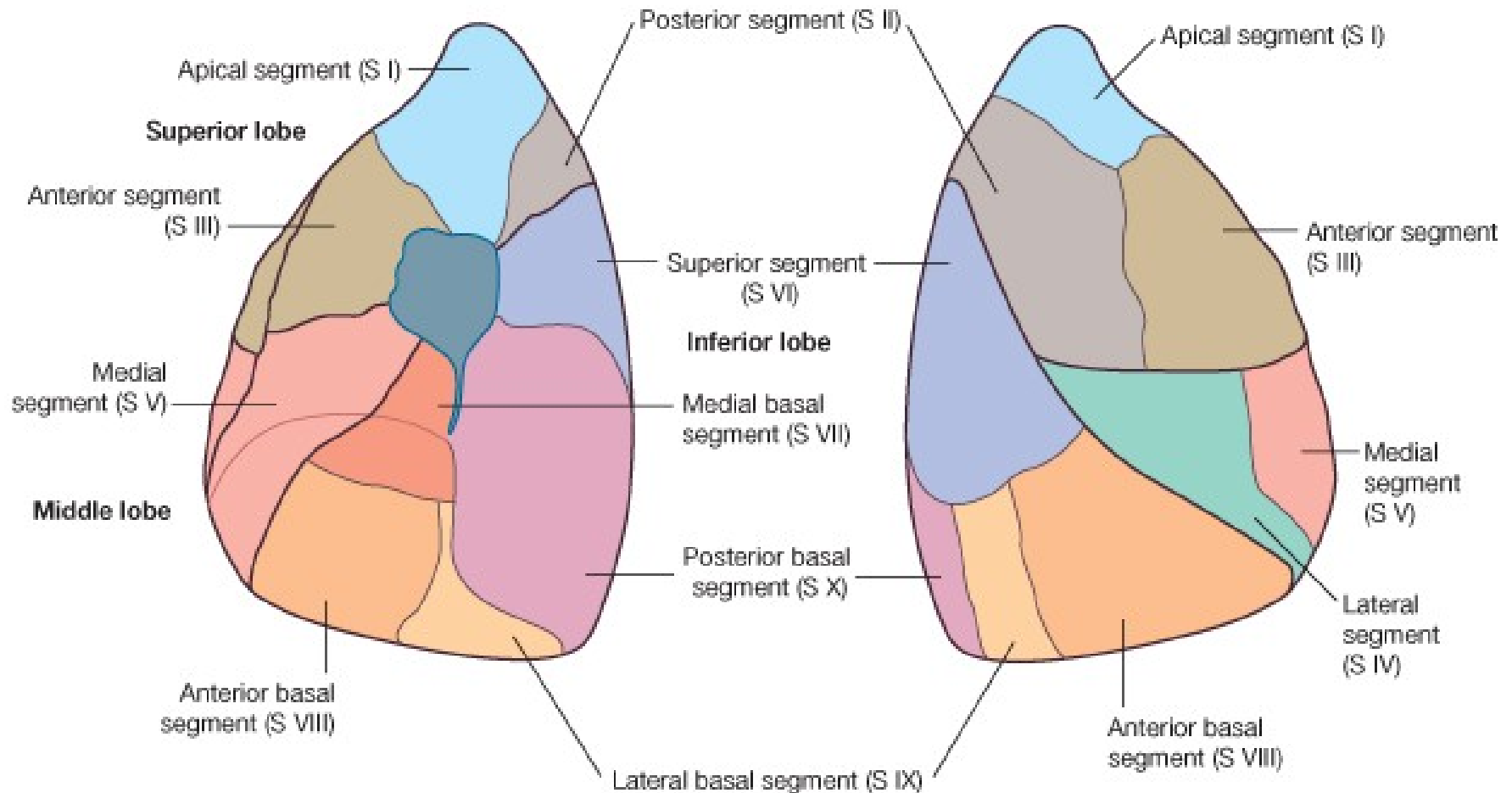
© Elsevier. Drake et al: Gray's Anatomy for Students - www.studentconsult.com

Bronchopulmonary segment: Right Lung

A

Medial view

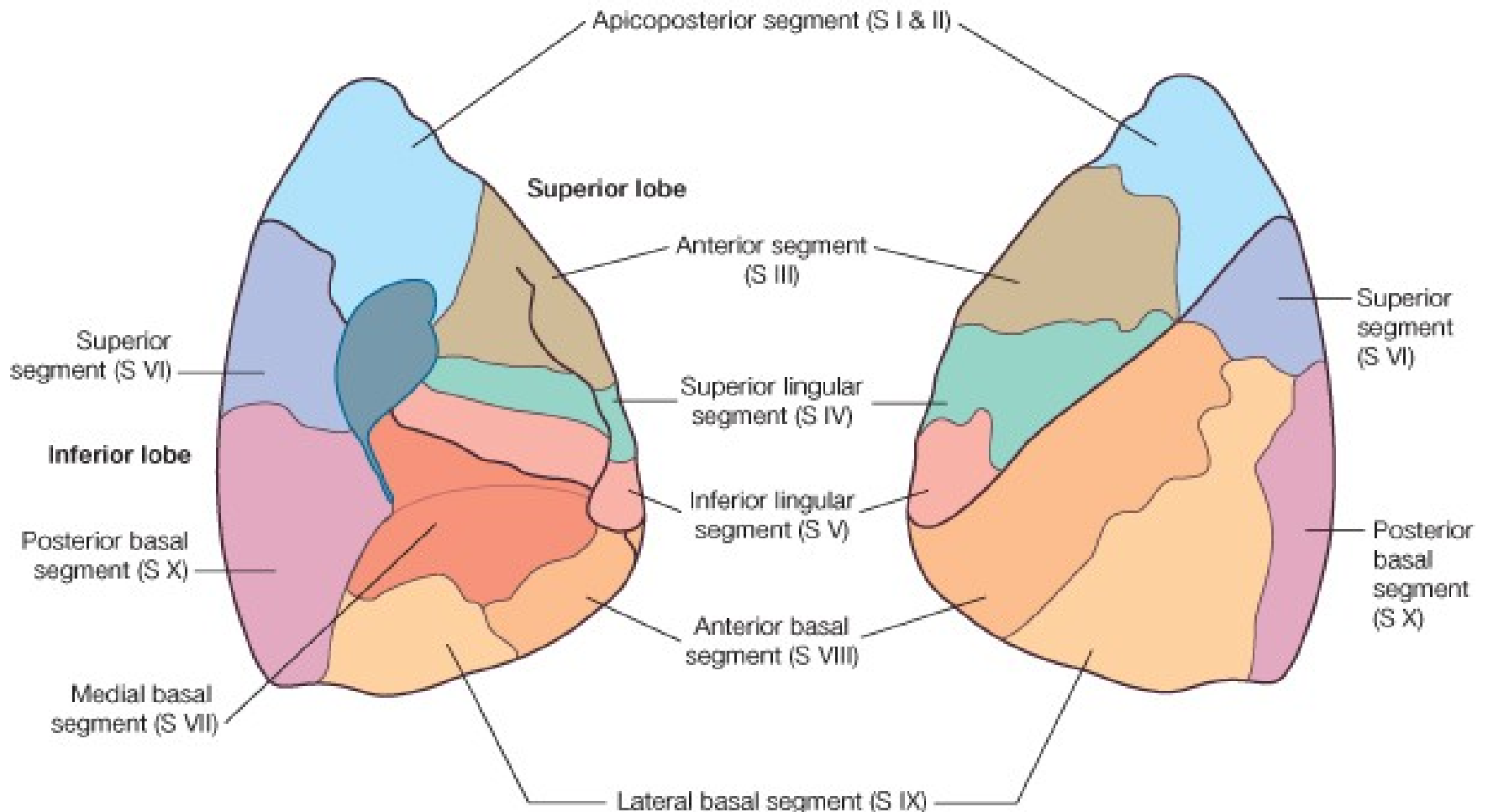
Lateral view



© Elsevier. Drake et al: Gray's Anatomy for Students - www.studentconsult.com

Bronchopulmonary segment: Left Lung

B



© Elsevier. Drake et al: Gray's Anatomy for Students - www.studentconsult.com

Bronchopulmonary segment

- **bronchopulmonary segment** is characterized by:
 1. Pyramidal in shape,
 2. surrounded by connective tissue,
 3. has segmental bronchus, segmental arteries, lymph vessels & autonomic nerves
 4. Segmental artery is in the segment, whereas segmental vein lies in the connective tissue b/n adjacent segments.
 5. A diseased segment can be removed surgically